

University of Tasmania Open Access Repository

Cover sheet

Title

Position analysis: Antarctic sea ice and climate change 2014

Author

Jan Lieser, Robert Massom, Alexander Fraser, Marcus Haward, Petra Heil, Delphine Lannuzel, Klaus Meiners, Jessica Melbourne-Thomas, Anthony Press, GD Williams

Bibliographic citation

Lieser, Jan; Massom, Robert; Fraser, Alexander; Haward, Marcus; Heil, Petra; Lannuzel, Delphine; et al. (2014). Position analysis: Antarctic sea ice and climate change 2014. University Of Tasmania. Report. <https://hdl.handle.net/102.100.100/495102>

Is published in:

Copyright information

This version of work is made accessible in the repository with the permission of the copyright holder/s under the following,

In Copyright.

Rights statement: © Copyright: The Antarctic Climate & Ecosystems Cooperative Research Centre 2014. This work is copyright. It may be reproduced in whole or in part for study or training purposes subject to the inclusion of an acknowledgement of the source, but not for commercial sale or use. Reproduction for purposes other than those listed above requires the written permission of the Antarctic Climate & Ecosystems Cooperative Research Centre.

If you believe that this work infringes copyright, please email details to: oa.repository@utas.edu.au

Downloaded from University of Tasmania Open Access Repository

Please do not remove this coversheet as it contains citation and copyright information.

University of Tasmania Open Access Repository

Library and Cultural Collections

University of Tasmania

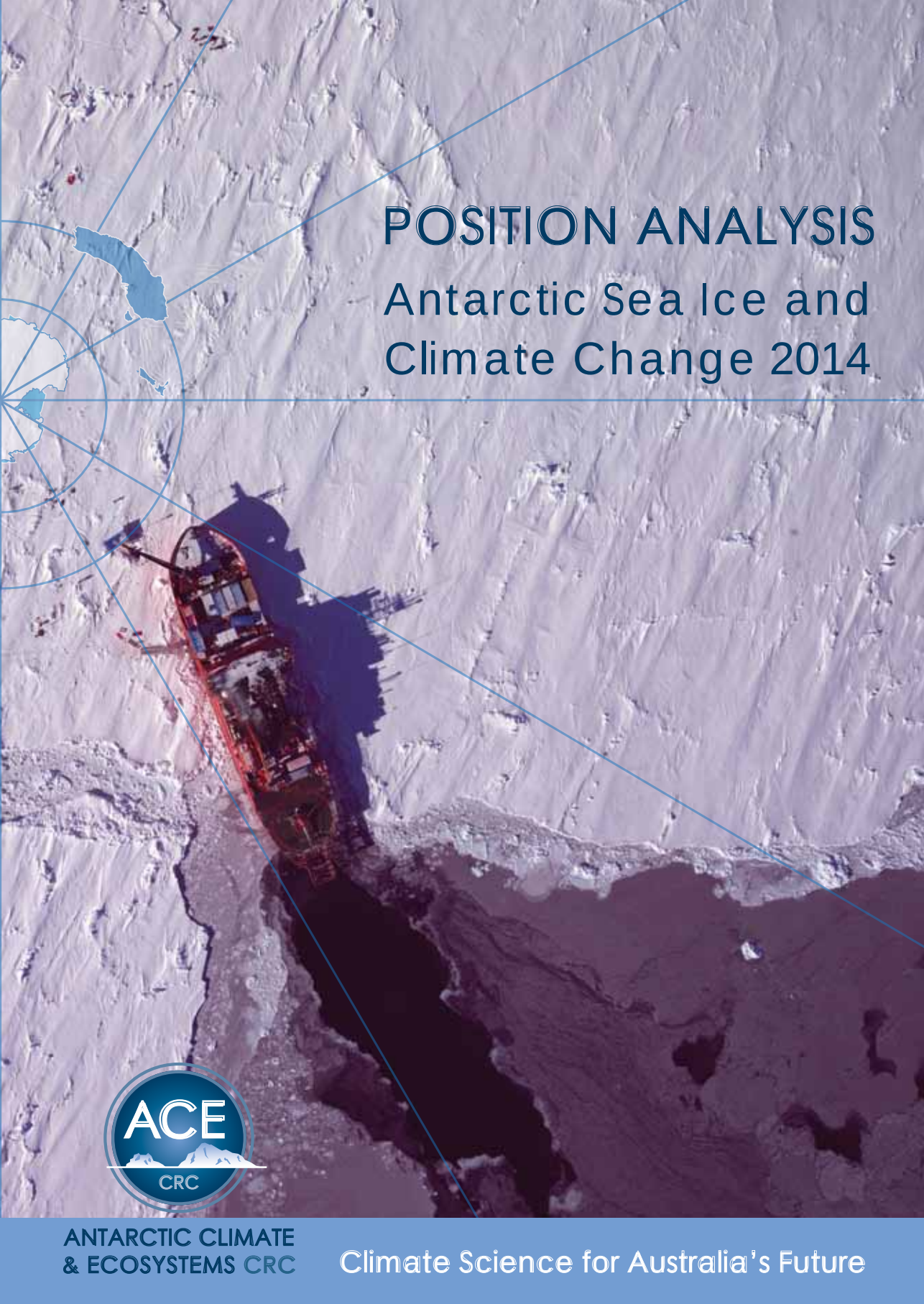
Private Bag 25

Hobart, TAS 7001 Australia

E oa.repository@utas.edu.au

CRICOS Provider Code 00586B | ABN 30 764 374 782

utas.edu.au

A satellite image of the Antarctic region, showing a large research ship (red and white) navigating through a field of sea ice. The ship is leaving a dark trail of open water behind it. In the upper left corner, there is a circular inset map showing the continent of Antarctica with a blue line indicating a specific location. The main title and subtitle are in the upper right, and the logos are in the bottom left.

POSITION ANALYSIS

Antarctic Sea Ice and Climate Change 2014



POSITION ANALYSIS: Antarctic Sea Ice And Climate Change 2014

PA07-131115

ISBN 978-0-646-91260-8

CONTACTS

Dr Jan L. Lieser

Marine Glaciologist

E: Jan.Lieser@acecrc.org.au

Professor Marcus Haward

Policy

E: Marcus.Haward@utas.edu.au

© Copyright: The Antarctic Climate & Ecosystems Cooperative Research Centre 2014.

This work is copyright. It may be reproduced in whole or in part for study or training purposes subject to the inclusion of an acknowledgement of the source, but not for commercial sale or use. Reproduction for purposes other than those listed above requires the written permission of the Antarctic Climate & Ecosystems Cooperative Research Centre.

Requests and enquiries concerning reproduction rights should be addressed to:

The Manager

Communications

Antarctic Climate & Ecosystems

Cooperative Research Centre

Private Bag 80

Hobart Tasmania 7001

Tel: +61 3 6226 7888

Fax: +61 3 6226 2440

Email: media@acecrc.org.au

www.acecrc.org.au

Cover image: Ice stations set up from RV Aurora Australis during the second Sea Ice Physics & Ecosystems experiment (SIPEX 2), East Antarctica, spring, 2012. Photo by Jan L. Lieser

The ACE CRC is a unique collaboration between core partners the Australian Antarctic Division, CSIRO, the University of Tasmania, the Australian Government's Department of Industry, the Alfred Wegener Institute for Polar and Marine Research (Germany), and the National Institute of Water and Atmospheric Research Ltd (New Zealand) and a consortium of supporting partners. It is funded by the Australian Government's Cooperative Research Centres Program.


An Australian Government Initiative




Australian Government
Department of the Environment
Australian Antarctic Division



ALFRED-WEGENER-INSTITUT
HELMHOLTZ-ZENTRUM FÜR POLAR-
UND MEERESFORSCHUNG



Printed on forestry stewardship certified paper

POSITION ANALYSIS:

Antarctic Sea Ice and Climate Change 2014

Contributing Authors

Jan L Lieser^{*1}, Robert A Massom^{1,2}, Alexander D Fraser¹, Marcus G Haward^{1,3}, Petra Heil^{1,2}, Delphine Lannuzel^{1,3}, Klaus M Meiners^{1,2}, Jessica Melbourne-Thomas^{1,2}, A J (Tony) Press¹, Guy D Williams¹

Affiliations

¹ Antarctic Climate & Ecosystems Cooperative Research Centre, University of Tasmania, Private Bag 80, Hobart, 7001, Australia.

² Australian Antarctic Division, Channel Highway, Kingston, Tasmania 7050, Australia.

³ Institute for Marine and Antarctic Studies, University of Tasmania, Private Bag 129, Hobart, Tasmania, 7001, Australia.

* coordinating author: Jan.Lieser@acecrc.org.au

This position analysis aims to:

1. Inform Australian governments and the community about our current state of knowledge about Antarctic sea ice;
2. Outline how sea ice influences, and responds to, global climate variability and change; and
3. Identify issues for consideration in policy development.

Citation: Lieser JL, Massom RA, Fraser AD, Haward MG, Heil P, Lannuzel D, Meiners KM, Melbourne-Thomas J, Press AJ & Williams GD (2013), Position analysis: Antarctic sea ice and climate change 2014, Antarctic Climate & Ecosystems Cooperative Research Centre, Hobart, Tasmania, Australia

GLOSSARY OF TERMS AND ACRONYMS USED

AADC – Australian Antarctic Data Centre;

Advection – Predominantly horizontal movement of a mass of fluid;

Albedo – The ratio of the reflected solar radiation to the incident solar radiation, usually referring either to the entire spectrum (broadband albedo) or just to the visible part of the spectrum;

Altimetry - Measurement of altitude using remote sensing technique, for example a radar or laser instrument;

Antarctic Bottom Water - Cold dense water that is formed near the Antarctic coast, flows down the continental slope to the abyssal plains and drives global ocean currents;

Antarctic Divergence - Atmospheric/oceanic boundary between the eastward-flowing Antarctic Circumpolar Current and the westward-flowing Antarctic Coastal Current;

Areal extent – The size of the area covered (→ sea ice extent);

ASAR – Advanced Synthetic Aperture Radar, an instrument on board the European ENVISAT satellite;

ATS – Antarctic Treaty System;

AUV – Autonomous Underwater Vehicle, an independent robot used for marine research;

AWI – Alfred Wegener Institute, Helmholtz Centre for Polar and Marine Research, Germany;

CCAMLR – Commission for the Conservation of Antarctic Marine Living Resources, part of the Antarctic Treaty System;

Convection – Heat transfer by mass motion of a fluid;

Coupled climate model – Numerical models of components of the climate system coupled together;

CPOM, UCL – Centre for Polar Observation and Modelling at University College London, United Kingdom;

CRYOS, EPFL – Laboratory of Cryospheric Sciences at École Polytechnique Fédérale de Lausanne, Switzerland;

CSIRO - Commonwealth Scientific and Industrial Research Organisation, Australia;

Dense Shelf Water – Dense water mass formed on the continental shelf from significant brine-rejection during enhanced sea ice formation, such as in → polynyas;

Dimethylsulfoniopropionate (DMSP) - Metabolite found in marine phytoplankton, seaweeds, and some species of aquatic vascular plants;

Ekman transport - Wind-driven transport of surface seawater, to the left of the wind direction in the Southern Hemisphere;

El-Niño/Southern Oscillation (ENSO) – A cycle of coherent and sometimes very strong year-to-year variations in sea-surface temperatures, convective rainfall, surface air pressure and atmospheric circulation that occur across the equatorial Pacific Ocean (source: NOAA);

Fast ice – Sea ice that has become "fastened" to the shore or between grounded icebergs;

Feedback – When a process triggers a change that in return influences the initial one;

Firn - Granular, partially consolidated snow that has commenced the transition to ice;

Freeboard – A measure of height of an ice (or snow) surface above a reference level, typically the sea level;

Hydrostatic equilibrium – A fluid is in hydrostatic equilibrium (or balance) when it is at rest, which means that external forces such as gravity and pressure are balanced;

Ice draft - Measurement of the sea ice thickness below the waterline;

Ice rafting – Horizontal pressure process whereby one piece of sea ice overrides another;

Ice ridging – Sea ice floes collide and pile up along a line, causing a steep-sloped ridge to rise up above the adjacent stretches of level ice, together with a much larger keel below sea level;

Ice shelf - A thick and extensive body of glacial ice attached to a coast and floating on the sea,

that gains mass by flow from the grounded continental ice sheet and by local snowfall;

Inter-annual - Time scale of more than one year but usually less than 10 years;

Inter-seasonal – Time scale between seasons;

IPCC AR5 – Intergovernmental Panel on Climate Change Fifth Assessment Report;

LiDAR - Light Detection And Ranging, one of the techniques applied for range measurements in altimetry;

Meridional exchange – North-south exchange (along a meridian);

Meridional overturning circulation – Global ocean circulation driven by temperature and salinity gradients (also referred to as → thermohaline circulation), which transports low-density waters to higher-latitudes in the upper layer and returns high-density waters equatorward at depth.

Mixed layer – Wind-mixed and brine-forced convection layer between the ocean surface and a depth usually ranging between 25 and 200 metres depending on the season;

NASA – National Aeronautics and Space Administration, United States;

NOAA - National Oceanic and Atmospheric Administration, United States;

Pack ice – Any area of sea ice that is not landfast;

Phenology - Timing of seasonal biological events;

Polynya - An anomalous region of open water or low ice concentration within the sea ice zone. Polynyas around the Antarctic coast are maintained where wind or ocean currents move ice away and are regions of enhanced ice production;

Ocean forcing – Physical constraints exerted by the ocean on the atmosphere or sea ice;

Outer pack – Northern part of the sea ice zone around Antarctica, connects to the marginal ice zone in the north and the inner pack (closer to the coast) in the south;

ROV – Remotely Operated Vehicle, a dependent robot operated through a tether;

Sea ice concentration - Proportion of the ocean surface area actually covered by sea ice in a given unit area, such as a satellite pixel. For example, 90% ice concentration signifies that 90% of the given area is covered with sea ice, with the remaining 10% being open water;

Sea ice extent - Area of the ocean covered by sea ice within the bounds of a specified minimum threshold ice concentration (typically taken to be 15%) in any given sector;

SeaWiFS - Sea-viewing Wide Field-of-view Sensor, the only scientific instrument on GeoEye's OrbView-2 (also known as SeaStar) satellite;

Snow ice – Ice layer at the interface between sea ice and the snow on top formed from snow and entrained seawater;

SIPEX 2 – The second Sea Ice Physics and Ecosystem eXperiment, an international multi-disciplinary research voyage, into the Antarctic sea ice zone in 2012 (the first SIPEX was in 2007);

Southern Annular Mode (SAM) - Pattern of climate variability between 40 °S and 65 °S;

Synthetic Aperture Radar (SAR) - A form of radar whose defining characteristic is its use of relative motion, between an antenna and its target region, to provide distinctive long-term coherent-signal variations, which are exploited to obtain finer spatial resolution than is possible with conventional beam-scanning means (source: Wikipedia);

Thermohaline circulation → meridional overturning circulation;

TLS - Terrestrial Laser Scanner;

WHOI – Woods Hole Oceanographic Institute, MA, United States.

EXECUTIVE SUMMARY

The annual expansion and contraction of sea ice in the Antarctic represents one of the biggest natural changes on Earth. At its maximum annual extent in September/October sea ice cover extends about 19 million square kilometres of the ocean around Antarctica – one and half times the size of the continent itself. In the summer sea ice shrinks to around 3 million square kilometres.

Antarctic sea ice plays a major role in the global climate system and in the ecology of the Southern Ocean. The sea ice region is the source of Antarctic Bottom Water, a major driver of global ocean overturning circulation. Sea ice is important in reflecting solar energy back into space. It is a habitat and a source of primary production for Southern Ocean ecosystems.

Sea ice cover in the Antarctic is changing. In the west Antarctic Peninsula region, sea ice extent has dramatically reduced, and the length of the sea ice 'season' has also shortened. In the Ross Sea region of the Antarctic, however, the maximum annual extent of sea ice cover has increased and its season extended. Overall, the maximum annual extent of Antarctic sea ice has increased by around 1.5% per decade since 1979 (or by around 285,000 square kilometres). This compares to a loss of 1.8 million square kilometres of sea ice maximum annual extent in the Arctic.

One important aspect of Antarctic sea ice that we know very little about is change to its volume. While satellites can measure the aerial extent of sea ice, reliable methods to remotely measure sea-ice thickness are only just emerging. In contrast, data from submarines in the Arctic have allowed detailed monitoring of regional changes to Arctic sea-ice thickness over a number of decades.

What is happening in the physical, chemical and biological systems beneath Antarctic sea ice is poorly understood. Because of the critical role of sea ice, it is vital that we develop a better understanding of the future trends in sea ice extent and volume, and consequences for Antarctic and Southern ocean ecosystems.

Australia is well placed to take a lead role in these studies - especially in the East Antarctic. Australia has a mature Antarctic research program that has demonstrated that it can collaborate with other nations, and across scientific disciplines, to do the complex science that is required to understand these important issues. Having the logistic and infrastructure capabilities will be vital to ensuring that this important region of the planet is understood.

WHAT IS SEA ICE

Sea ice is frozen seawater that forms when the surface of the ocean cools to the point of freezing (at approximately -1.8°C). Every winter, extensive freezing of the Southern Ocean forms a vast sea ice cover that surrounds the Antarctic continent - to drastically modify the properties of the ocean surface and the interaction between two key elements of the global climate system, namely the ocean and atmosphere. As its density is less than that of seawater, sea ice floats on the ocean surface, where it forms a heterogeneous matrix consisting mainly of ice and small pockets (cells) of brine [Petrich and Eicken, 2010], and its characteristics are modified by physical, chemical and biological processes.

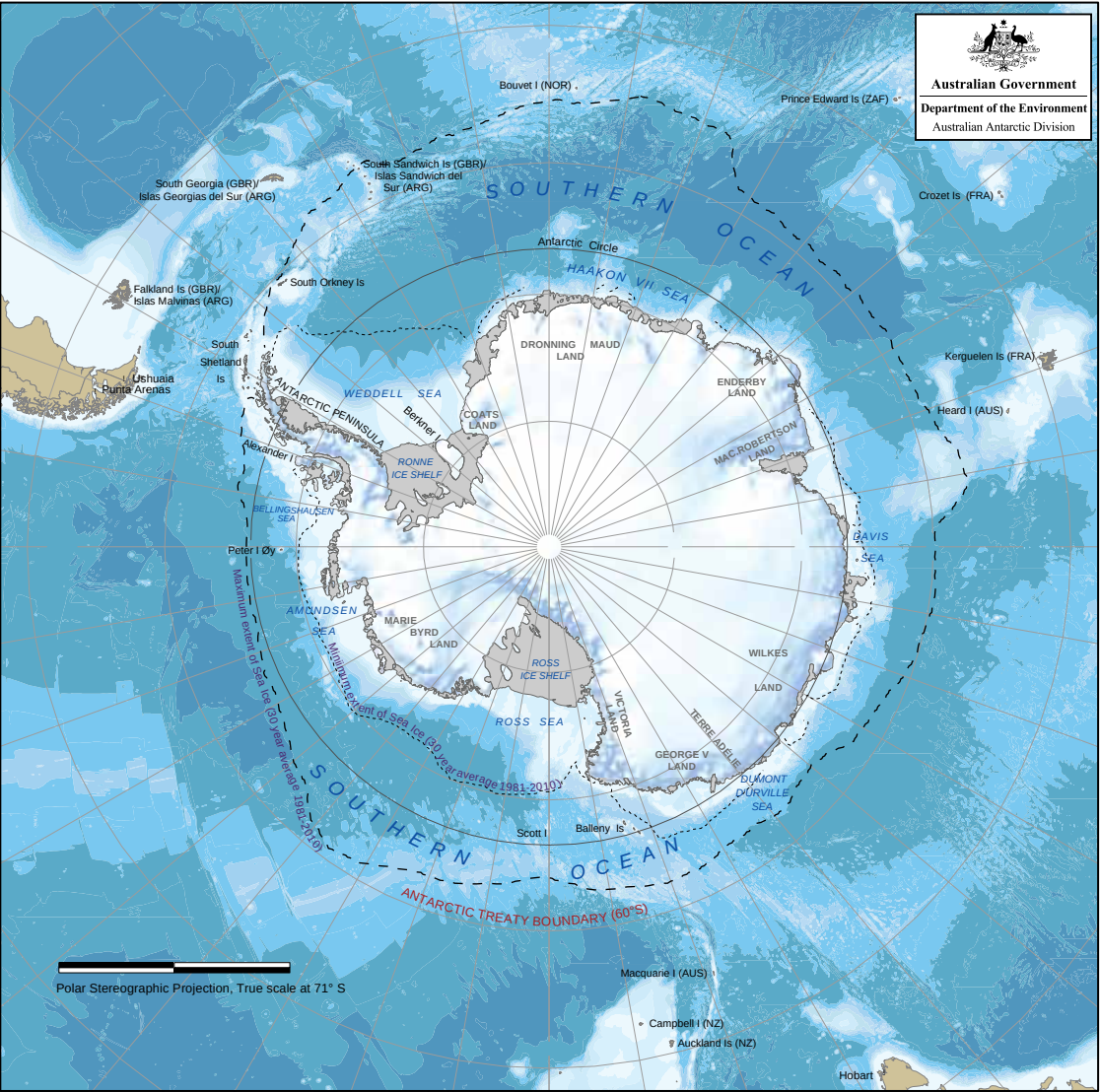


Figure 1. Maximum and minimum extent of Antarctic sea ice (30-year average for 1981-2010).

Areal extent and seasonality

At its maximum annual extent in September/October and as depicted in Figure 1, this sea ice cover extends over an extraordinary ~19 million square kilometres of the Southern Ocean [Cavalieri and Parkinson, 2008; Comiso, 2010]; by comparison, the area of the Antarctic continent is ~14 million square kilometres, while that of Australia is ~7.7 million square kilometres. During summer, this areal extent shrinks to ~3 million square kilometres (in February). As described below, this seasonal expansion and contraction is of immense climatic, biological and biogeochemical importance, and it follows that any significant changes in sea ice areal extent and seasonality (and other characteristics) have wide-ranging and indeed global ramifications. Whereas areal extent is a measure of the area of the ocean surface covered by sea ice at any given time, seasonality collectively describes the timings of annual sea ice advance and retreat and the resultant duration of sea ice coverage for a particular location.

The width of the sea ice zone at maximum extent varies substantially around the continent – from up to ~2200 km in the Weddell Sea sector (where it occupies a deep poleward embayment) to only a few hundred kilometres in some places, notably between 120° E and 135° E. In this region, the coast extends further north than in other regions (with the exception of the roughly north-south trending Antarctic Peninsula).

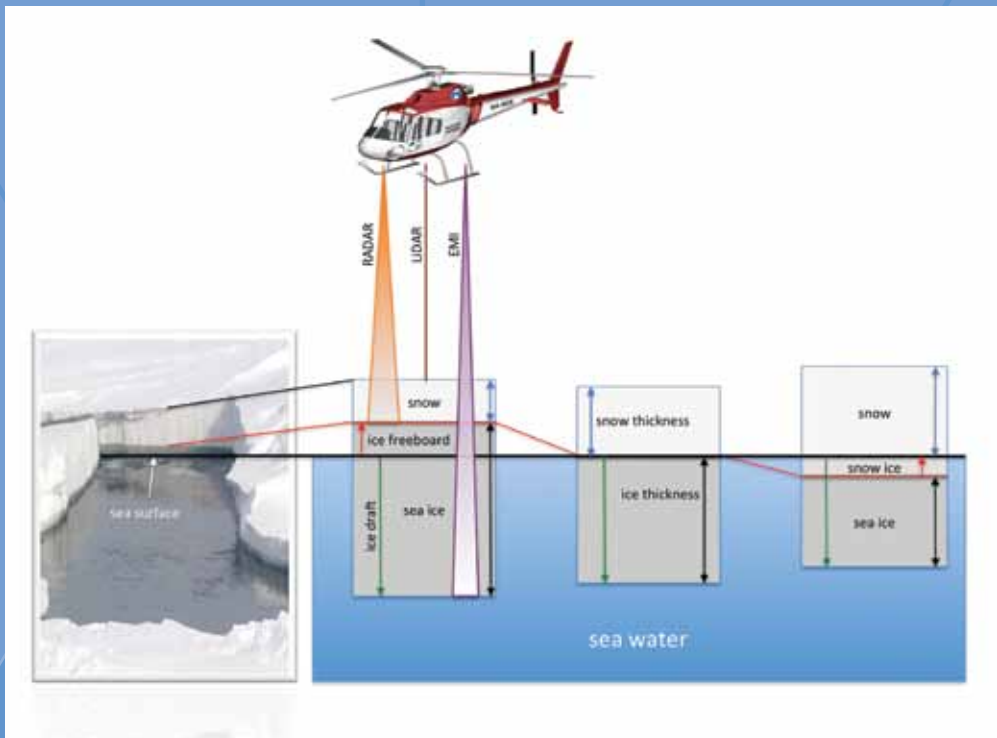
How does sea ice form?

The first stage in sea ice development is the formation of individual crystals comprising fine spicules or platelets known as *frazil* ice. With further freezing, frazil crystals coagulate into a soupy layer on the ocean surface known as *grease ice*. Subsequent development depends on whether calm or stormy conditions prevail. Under calm conditions, frazil and grease ice tend to consolidate into thin flexible sheets called *nilas*, which thicken over time. Under stormier conditions, a widespread process of sea ice development around Antarctica is the *pancake cycle*. By this process and under the influence of wind and wave action, frazil crystals coagulate to form small “pancakes” – or small floes with upturned edges - on the ocean surface (a *floe* is defined as a contiguous piece of ice on the water surface). These pancakes subsequently bond together and raft (over-ride by colliding with each other), to eventually form larger floes or a consolidated ice cover. Ice growth also occurs on the underside of existing sea ice as heat is conducted upwards from the ice-ocean interface and through the ice, that is by *thermodynamic* growth. This type of ice is called congelation or columnar ice.

In general, the accumulation of snow on sea ice older than a few days plays a key role in determining the properties and development of the underlying ice, due to its strong insulating characteristics and high albedo [Sturm and Massom, 2010]. This in turn modifies the overall climatic, biological and biogeochemical function of the ice.

Importantly, snow also makes a significant contribution to Antarctic sea ice formation via the formation of snow ice [Maksym and Markus, 2008]. This widespread phenomenon largely occurs when the weight of the snow is sufficient to depress the ice surface below sea level. The resultant influx of seawater saturates the lower snow layer, which may

UNDERSTANDING HOW SEA ICE THICKNESS IS MEASURED



The sea surface is indicated by the thick black line as a reference surface. Sea ice (grey) floats at the surface of the ocean with a snow cover on top (white). Black arrows show the ice thickness, blue arrows snow thickness; the combination of both is referred to as total thickness. The ice freeboard is indicated by the red line and typically sits above the sea surface (at the interface between the sea ice and the snow), but with sufficient snow load can be suppressed below the sea surface, which then leads to snow ice formation (light grey). The sum of snow thickness and ice freeboard is called total freeboard. Ice draft is measured below the surface (green arrows). Different remote sensing techniques can 'see' different surfaces. LiDAR reflects from the snow surface (or ice surface in case of no snow cover) and therefore gives an estimate of snow freeboard. The RADAR penetrates the snow cover and gives an estimate of snow thickness. Electromagnetic Induction (EMI) systems give an estimate of total thickness in combination with LiDAR. Space-borne remote sensing techniques (LiDAR or RADAR) provide measurements of ice or snow freeboard, from which ice thickness can be computed if ice, snow and water density and snow thickness are known.

subsequently freeze to form a type of salty and granular sea ice that accretes to the surface of the existing ice [Jeffries *et al.*, 2001].

How old is the sea ice?

In contrast to the continental ice mass covering Antarctica, which has built up over hundreds of thousands of years of accumulated snowfall, most of the sea ice around Antarctica is seasonal and therefore less than one year old - so-called *first-year ice*. This type of ice has a thickness range of 0.3-2.0 m (typically), although greater thicknesses can occur by ice deformation.

Perennial sea ice (ice that survives one or more summer melt seasons) is generally thicker but is largely confined to the Weddell, Amundsen and eastern Ross seas (Figure 1). These are not only some of the southernmost regions around the continent, but they are also areas where ice is pushed by winds and ocean currents toward the coast [Comiso and Nishio, 2008].

Ice motion and deformation

Rather than forming a static, unbroken veneer on the ocean surface, most of the sea ice around Antarctica comprises a highly-mobile aggregation of floes that constantly move with the wind and currents. This is termed *pack ice*. Due to the highly-dynamic response of the pack ice zone to the passage of storms, its characteristics also change constantly. New areas of open water between floes (termed *leads*) freeze rapidly under winter conditions to form areas of new ice, whereas existing ice thickens not only thermodynamically (via heat loss to the atmosphere) but also *dynamically* as floes are pushed together and deform under convergent conditions. Moreover, the characteristics of the outer part of the pack, called the *marginal ice zone*, are strongly affected by ocean waves that not only generate new pancake ice formation but also break up existing floes.

Across the greater Antarctic pack, sea ice growth by thermodynamic processes alone is self-limiting as the ice thickens to insulate the ocean surface from the cold atmosphere. Ice motion and deformation then play a dominant role in thickening the ice beyond ~0.5 m - ice seldom attains a thickness of >0.1-0.2 m before being rafted or piled into pressure ridges by the action of wind, waves, ocean currents and tides [Allison *et al.*, 1993; Worby *et al.*, 1998]. Such is the power of these mechanical deformation processes that pressure ridges can be up to ten or more metres thick in regions of first-year ice [Massom *et al.*, 2006; Williams *et al.*, 2013].

On horizontal scales of tens of metres to tens of kilometres, the net result of this complex interplay of thermodynamic (freeze/melt) and dynamic (deformation) processes is a complicated mixture of different types, ages and thicknesses of ice. On the broader scale of hundreds of kilometres, however, regions or zones with similar characteristics become apparent [Massom *et al.*, 1999; Massom and Stammerjohn, 2010].

Sea ice motion is also a key factor in transporting sea ice (and its snow cover) from one region to another, that is from an area of formation to an area of melt, to affect the freshwater budget of the high-latitude Southern Ocean.

As such, ice drift is one of the most important features of the Antarctic sea ice zone. As discussed later, there are indications that regional patterns of ice drift are changing around Antarctica [Holland and Kwok, 2012], with major implications for ice concentration, extent, thickness, age and seasonality. *Sea ice concentration* is defined as the percentage coverage of sea ice (versus open ocean) within a given ocean area.

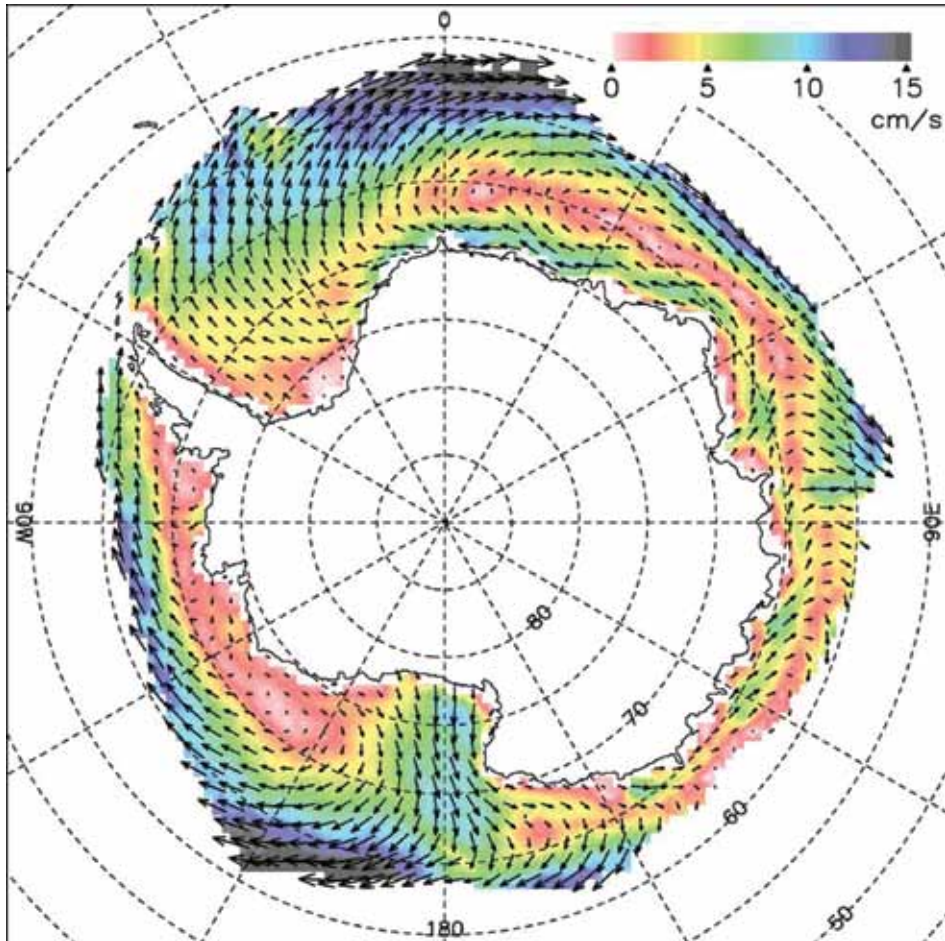
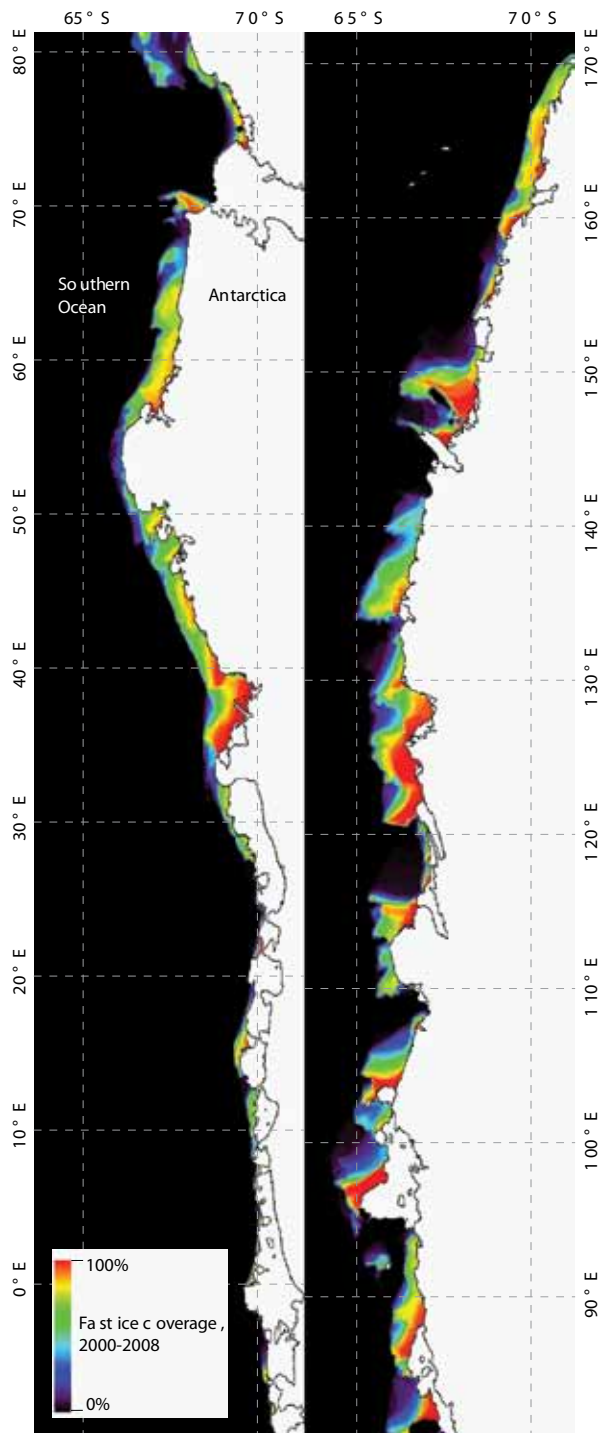


Figure 2. Mean sea ice motion in the Southern Ocean averaged over a 10-year period (1992-2001), with ice drift speeds indicated by lengths of vectors and the background colours. This large-scale map of ice motion was derived from analysis of satellite data. From Kimura [2004].



As a background against which to discuss recent ice motion change (in Section 4), Figure 2 shows mean Antarctic sea ice motion over a 10-year period (derived from satellite data). Being a long-term average, these ice-drift patterns strongly reflect climatological wind and ocean surface current patterns. Outstanding features are the circumpolar band of mean eastward ice motion centred on $\sim 60^\circ$ S in the Atlantic sector and $\sim 65^\circ$ S in the Pacific sector, and a narrow westward-flowing coastal current that is predominantly in the Atlantic and Indian sectors. These two regimes, namely the *Antarctic Circumpolar Current* to the north and *Antarctic Coastal Current* to the south, are separated along the *Antarctic Divergence* [Heil and Allison, 1999], but are connected in certain regions by northward retroflexions, for example along 85° E [Rintoul et al., 2008]. Other key features are major gyre systems in the Weddell and Ross seas and the Prydz Bay region of East Antarctica.

Figure 3. A projection of the East Antarctic coast, showing the mean percentage of time (per year) with fast ice coverage from 2000 to 2008 (derived from satellite data). A figure of 100% indicates fast ice that persists throughout each year (that is multi-year fast ice, which can attain considerable thickness). From Fraser et al. [2012].

Fast ice

Not all sea ice is constantly moving in response to winds and currents. A relatively narrow (up to ~200 km wide) but generally consolidated band of stationary or landfast sea ice (fast ice) occurs around the coastal margins of the continent (Figure 3), where it is held in place by coastal promontories, embayments and/or grounded icebergs [Giles *et al.*, 2008]. Fast ice can form and break out annually [Heil, 2006; Fraser *et al.*, 2012], or can attain great ages and thicknesses in regions where the ice is more tightly locked to the coastal margins [Massom *et al.*, 2010].

THE ROLE OF ANTARCTIC SEA ICE IN THE CLIMATE SYSTEM

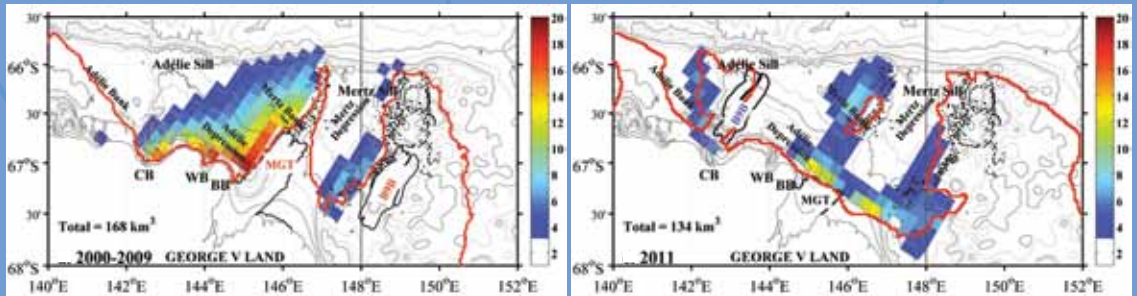
Why is sea ice important?

Antarctic sea ice forms a relatively thin but seasonally-extensive and highly-dynamic cover on the surface of the high-latitude Southern Ocean that is intimately tied to patterns of atmospheric and oceanic temperature and circulation. As a result, it is one of the most sensitive indicators of climate change and variability on Earth. Crucially, the ice cover is not only affected by a changing and variable environment, it also plays an active though poorly-understood role in modulating such change and variability via its influence on the ocean and atmosphere. Thus the sea ice, ocean and atmosphere form a complex interactive system. On shorter time scales, it is intimately tied to the weather, and responds rapidly to changes in wind direction and strength associated with the passage of synoptic systems. In this way, the ice edge location can vary by tens of kilometres from one day to the next.

The climatic significance of Antarctic sea ice lies in the major modifying effect it has on atmosphere-ocean interaction processes and the surface radiation budget, with the annual cycle of areal expansion and contraction constituting one of the largest seasonal geophysical changes on the surface of the planet. Primarily, the ice acts as an insulating blanket between the cold atmosphere and relatively warm ocean, and as a physical barrier to the exchange of moisture, gases and momentum between the two. Being white, sea ice also reflects substantially more sunlight back into space than does open ocean, which is dark and therefore absorbs more sunlight and heats up. This high albedo (reflectivity) characteristic of sea ice is important for climate feedback and amplification processes [Perovich *et al.*, 2008]. These effects are strongly modified by the presence of substantial accumulations of snow on the surface of Antarctic sea ice, given the even greater insulating and reflective properties of the snow [Sturm and Massom, 2010]. In general, the degree to which sea ice affects ocean-atmosphere interactions depends on the ice extent, its concentration and its thickness distribution.

Antarctic sea ice also plays a key role in ocean water mass modification and formation. As it forms and grows, the ice expels salt into the underlying ocean as brine, to increase the salinity and density of the water column [Toggweiler and Samuels, 1995]. Conversely and in spring, sea ice melt releases fresher water, to form a stable low-salinity ocean surface layer.

WHAT HAS THE MERTZ GLACIER CALVING SHOWN US?



Sea ice production in coastal polynyas, which is critical to the formation of Antarctic Bottom Water (AABW), has declined significantly off George V Land in East Antarctica since the calving of the Mertz Glacier Tongue in February 2010. From *Tamura et al. [2012]*.

Scientists from the ACE CRC used satellite data to detect regions of thin ice and attribute the associated heat flux to estimate sea ice production (colour shaded above in metres per year). A 14 – 20% decrease (relative to the mean from 2000—2009) was detected in sea ice production off George V Land in the two winters after the glacier tongue's calving and the associated realignment of vast iceberg B9B [*Tamura et al., 2012*]. These findings suggest that 50% of this decrease in sea ice production can be attributed to the changes in the icescape post-calving (thick red line) impacting the distribution and strength of polynya activity. The Mertz Glacier Polynya is important to global climate as a site of significant AABW formation [*Rintoul, 1998*].

Another ACE CRC study, by *Shadwick et al. [2013]*, has highlighted strong links between this abrupt natural event and biological productivity and carbon drawdown in the region. Specifically, breakout and melt of thick multi-year fast ice released by the movement of iceberg B9B and the Mertz Glacier Tongue led to significant input of meltwater into the ocean surface layers. This likely enhanced the availability of light and iron, supporting a diatom bloom that doubled carbon uptake relative to pre-calving conditions.

Particularly high rates of ocean-atmosphere heat loss and resultant ice formation and brine rejection occur in recurrent and persistent areas of open water and thin ice at certain locations around the Antarctic coast known as *polynyas*. Within these regions, sea ice is swept away by strong winds and currents as quickly as it forms. In this way, polynyas act as “ice factories” for the wider sea-ice zone. This enhanced sea-ice formation and corresponding salt rejection forms cold, saline *Dense Shelf Water* that in discrete locations can be transported northward across the shelf break, with sufficient density to mix down the continental slope and produce *Antarctic Bottom Water (AABW)*. This AABW in turn plays a key role in driving global ocean thermohaline circulation. As such, Antarctic coastal polynyas in the southern Weddell and Ross seas and adjacent to the Mertz Glacier tongue and Cape Darnley in East Antarctica are of global significance in spite of being relatively small (10s to 100s of kilometres across). In this way, the influence of sea ice reaches the ocean abyss as well as the tropics [Rintoul *et al.*, 2001]. Moreover, *High-Salinity Shelf Water* associated with sea ice formation controls circulation and melt processes under ice shelves [Galton-Fenzi *et al.*, 2012]. It follows that any lasting change to polynya systems could have far-reaching climatic consequences.

Although fast ice comprises a relatively small fraction of overall sea ice area – it varies seasonally between 5% (in winter) and 35% (in summer) for East Antarctica (Figure 3) [Fraser *et al.*, 2012] - it is important for a number of key reasons. For one thing, it too is a sensitive indicator of climate change and variability around the Antarctic coastal margins [Murphy *et al.*, 1995; Heil, 2006]. It can also attain a considerable thickness, of >5 m in places [Massom *et al.*, 2010], and thus represents a significant freshwater store. Moreover, fast ice plays an indirect role in global thermohaline ocean circulation by forming a major component in coastal polynya systems [Tamura *et al.* 2012], and it may in certain regions around the Antarctic coast act to mechanically stabilise floating glacier tongues and ice shelves [Massom *et al.*, 2010]. Last but not least, its presence can both impede and facilitate logistical access to Antarctic bases.

In addition to its important physical role, sea ice dominates the physical and chemical seasonal dynamics of the high-latitude Southern Ocean and plays a crucial role in marine ecosystems [Smetacek and Nicol, 2005; Thomas and Dieckmann, 2010]. Around the coast, fast ice forms a crucial breeding platform for Emperor penguins (*Aptenodytes forsteri*) and Weddell seals (*Leptonychotes weddelli*) [Kooyman and Burns, 1999], and its breakup and the resultant release of nutrients and freshwater into the water column has been associated with algal blooms [Shadwick *et al.*, 2013].

Important changes

In the past few years, an increasing number of headlines have reported dramatic changes occurring in the high latitudes (both north and south) - with Arctic summertime (minimum annual) sea ice extent reaching a record minimum in 2012 since satellite records began in 1979 [Simmonds and Rudeva, 2012] and ice shelves disintegrating along the Antarctic Peninsula [Cook and Vaughan, 2010]. This indicates that climate change is occurring already, and in a way that was not foreseen only a few years ago.

In response to a warming climate, annual mean Antarctic sea ice extent and volume is forecast to reduce by 24% and 34%, respectively, by 2100 [Arzel *et al.*, 2006], with the most pronounced reduction being at the end of summer. This could have far-reaching effects on global climate and marine ecosystems. It would also allow greater access to Antarctic waters for both commercial and research vessels.

RECENT CHANGES IN ANTARCTIC SEA ICE

Sea ice extent: Antarctic versus Arctic

Since 1979, when reliable sea ice data became available from satellite sensors, the two polar regions have experienced markedly different trends in the overall (circumpolar) extent of sea ice coverage. In the Arctic, the extraordinary decline in annual sea ice extent - of 3.5-4.1% per decade (95% confidence level) for the period 1979-2012 [IPCC, 2013] - is one of the most dramatic and conspicuous examples of climate change anywhere on Earth [Perovich, 2011]. The rate of sea ice loss in the Arctic, which is particularly strong in summer, is even more rapid than that predicted by models [Stroeve *et al.*, 2007], and has initiated strong climate feedbacks that have enhanced the warming in the region [Screen and Simmonds, 2010].

By contrast and as depicted in Figure 4a, annual sea ice extent around Antarctica has shown a small increasing trend of 1.2-1.8% per decade (around 95% confidence level) for 1979 to 2012 [IPCC, 2013]. The cause of the contrasting responses of the two polar sea-ice covers is subject to lively debate, but is thought to relate to differences in their geographical settings, respective sea ice growth and decay processes, large-scale climate interactions, and ice-ocean interactions and feedbacks [Turner and Overland, 2009; Maksym *et al.*, 2012].

The slight expansion observed in overall Antarctic sea-ice areal extent in fact masks dramatic regionally-contrasting changes that have occurred in both the extent and seasonality of sea ice coverage [Comiso *et al.*, 2011; Stammerjohn *et al.*, 2012]. For ice extent, these regional differences in trends are shown in Figure 4. Most notably, sea ice extent has increased in the Ross Sea sector by $5.0 \pm 0.6\%$ per decade (for the period 1979-2008), while major loss has occurred in the Bellingshausen/Amundsen seas sector (at a rate of $-7.1\% \pm 0.9\%$ per decade). By comparison, trends in the remaining Antarctic sectors are similar to the overall circumpolar trend.

As such, the observed trend towards increased Antarctic sea ice extent overall is in fact the sum of similar trends in East Antarctica and much larger and opposing trends in West Antarctica. A number of hypotheses have been proposed; all suggest that the response of Antarctic sea ice to climate change (and variability) is far from simple. A recent satellite-based study, for example, suggests that wind-driven changes in sea ice advection are a dominant driver of trends around much of West Antarctica, whereas wind-driven thermodynamic changes dominate elsewhere [Holland and Kwok, 2012]. Other studies have proposed that changes in atmospheric temperature and wind stress [for example Liu *et al.*, 2004; Turner *et al.*, 2009; Lefebvre and Goosse, 2008] together with

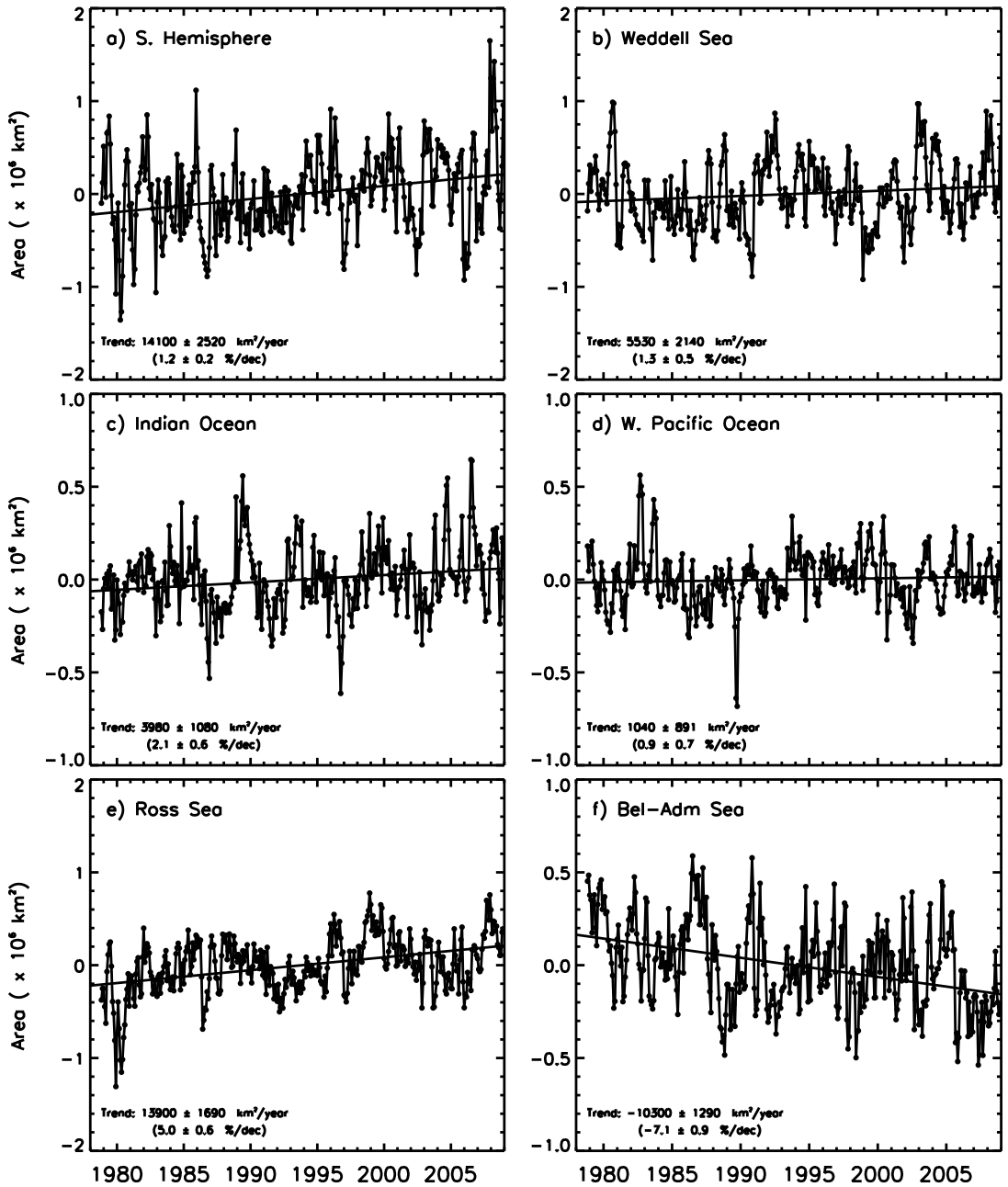


Figure 4. Trends in monthly ice extent, derived from satellite data and expressed as anomalies against the long-term mean (1978-2008), in the (a) entire Southern Ocean, (b) Weddell Sea, (c) Indian Ocean, (d) Western Pacific Ocean, (e) Ross Sea, and (f) Bellingshausen/Amundsen seas. From *Comiso et al.*, 2011. Locations are marked on Figure 1.

changes in precipitation [Liu and Curry, 2010], ocean temperature [Jacobs and Comiso, 1997; Meredith and King, 2005], and atmosphere and/or ocean feedbacks [Zhang, 2007; Stammerjohn et al., 2012] are responsible. There may also be an important link to the ice sheet. Bintanja et al. [2013] propose that enhanced input of meltwater from basal melting of Antarctic ice shelves (Pritchard et al. [2012]) into the upper ocean leads to the formation of a relatively cool and fresh surface layer that shields the surface ocean from the warmer deeper waters. These surface layers can then freeze more easily, which could explain why observed overall Antarctic sea-ice extent trends peak in autumn and early winter.

Underpinning the changes and regional contrasts outlined above are large-scale patterns of atmospheric forcing associated with the Southern Annular Mode (SAM) and to El Niño-Southern Oscillation (ENSO)-related teleconnections across the southern Pacific Ocean [for example Renwick, 2002; Yuan and Li, 2008; Stammerjohn et al., 2008; Turner et al., 2009; Pezza et al., 2012]. Crucially, the dominant circumpolar westerly wind belt over the Southern Ocean has strengthened by 15-20% and has migrated polewards by 1-2° of latitude since the late 1970s [Korhonen et al., 2010; Turner and Marshall, 2011]. This change has been linked to a change in the atmospheric pressure gradient in the Southern Hemisphere: pressure has increased at mid-latitudes but decreased around the Antarctic coast [Thompson et al., 2011]. The SAM, which is a measure of this gradient and is the dominant mode of variability in atmospheric circulation at high southern latitudes [Simmonds, 2003], has changed to a more positive phase in the austral summer and autumn over the past half century. This change has been linked to ozone depletion [Gillett et al., 2009; Thompson et al., 2011] and increased greenhouse gas concentrations in the atmosphere [Marshall et al., 2004]. It has been noted elsewhere that change and variability in key large-scale climate modes such as SAM and ENSO could also arise through natural variability [Holland and Kwok, 2012].

A key factor affecting the dramatic changes in sea ice extent in the West Antarctic sector has been the deepening of the Amundsen Sea Low (ASL), which is a recurrent low atmospheric pressure anomaly in the South Pacific, due to the increased intensity of westerly winds and the geographical configuration of the Antarctic continent [Turner et al., 2013a]. Resultant stronger and/or more persistent southerly (colder) winds along the western flank of the ASL have generated more extensive sea ice in the Ross Sea, while northerly (warmer) winds in the Bellingshausen Sea have led to less extensive ice there. Proxy information on historical sea ice extent which can be obtained from ice cores from the Antarctic Ice Sheet [Curran et al., 2003] and sources such as whaling data [de la Mare 1997, 2008] are crucial in terms of extending the satellite data record back in time (before the 1970s) - to place recent change/variability into longer-term context. For example, ice core data from the Antarctic Peninsula suggest that sea ice decline in the Bellingshausen Sea since 1979 is part of a trend that has persisted since the early 1900s due to a progressive deepening of the Amundsen Sea Low [Abram et al., 2010].

Sea ice seasonality

In addition to sea ice extent, changes in sea-ice seasonality are crucially important for a

number of reasons [Massom and Stammerjohn, 2010; Maksym et al., 2012]. Ice-free summer duration controls solar heating and wind-mixing of the upper ocean [Montes-Hugo et al., 2009; Perovich et al., 2008], thus affecting sea-surface temperatures [Meredith and King, 2005] and ocean upwelling [Martinson, 2012]. Moreover, the life cycles of high-latitude marine organisms (from micro-organisms to whales) are specifically adapted to sea ice and its seasonal rhythms [Thomas and Dieckmann, 2010], and serious consequences can arise when these rhythms change (see Section 4).

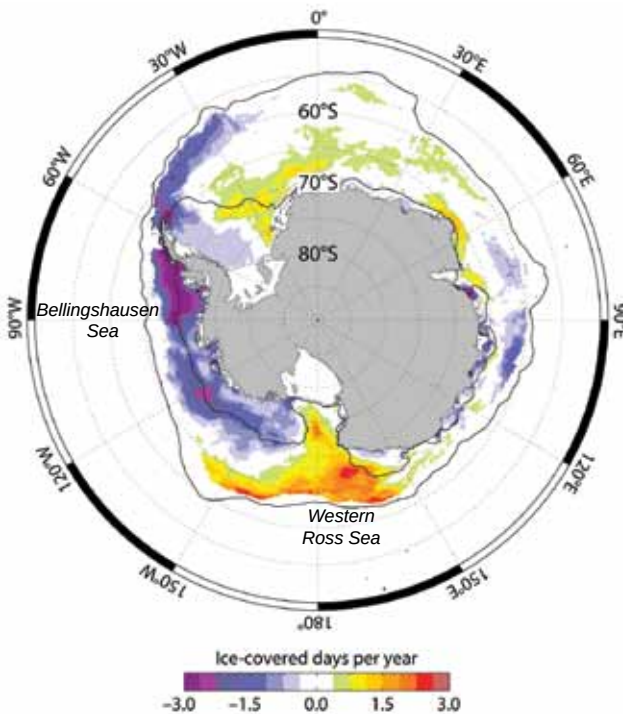


Figure 5. Trends in the length of the Antarctic sea ice-covered season (ice duration) for the period 1979/80-2010/11. These are derived from daily satellite ice extent records (1979-present), which have been used to determine, for each sea ice year (February to February) and each satellite pixel location, the day sea ice advanced to that location and the day it retreated from that location. Methods are described in Stammerjohn et al. [2008], after Parkinson [2002]. The black lines denote the mean minimum and maximum ice extents for the same time period. From Maksym et al. [2012].

Strongly contrasting regional changes have also occurred in timings of annual sea-ice advance and retreat and resultant sea ice duration around Antarctica (Figure 5). In contrast to the Arctic, where most regions show trends towards shorter sea ice duration, patterns of regional-scale change in Antarctic sea ice seasonality are accentuated by two regions in particular [Simpkins et al., 2013; Stammerjohn et al., 2012]. These are the Antarctic Peninsula-Bellingshausen Sea (AP-BS) and Western Ross Sea (WRS) regions. Of major concern from both a physical and biological perspective is a substantial shortening of the sea ice season (lengthening of the open water season) in the AP-BS region – of 3.3 months over the period 1979/80 to 2010/11. This extraordinary change results from later advance (by ~2 months) and earlier retreat (by ~1.3 months). In fact, the rate of increase in ice-free conditions in the Bellingshausen Sea region (of 3.1 ± 1.0 days per year as depicted in Figure 5) is even greater than that which is occurring in regions of greatest ice loss in the Arctic [Stammerjohn et al., 2012].

In stark contrast, the annual sea ice season in the WRS region has significantly lengthened, by ~2.6 months over the period 1979/80 to 2010/11. Here, earlier sea ice advance (by ~1.4 months) is followed by a later retreat (by ~1.2 months).

By comparison, and as shown in Figure 6, patterns of change in the relatively narrow East Antarctic sea-ice zone are far more complex, comprising mixed signals on regional to local scales [Massom *et al.*, 2013]. Indeed, pockets of strongly positive and negative trends in ice season duration (of ± 2 -3 days per year) occur in near juxtaposition in certain regions, for example Prydz Bay. A negative trend in sea ice duration (of -1 to -3 days per annum) occurs in fairly isolated pockets in the outer sea ice zone at $\sim 60^\circ$ S to 62° S and $\sim 95^\circ$ E to 110° E, and in various near-coastal areas, including an area of particularly strong and persistent change near Davis Station and between the Amery and West Ice Shelves. These areas are largely associated with coastal polynyas that are regionally-important as sites of enhanced sea ice production/melt.

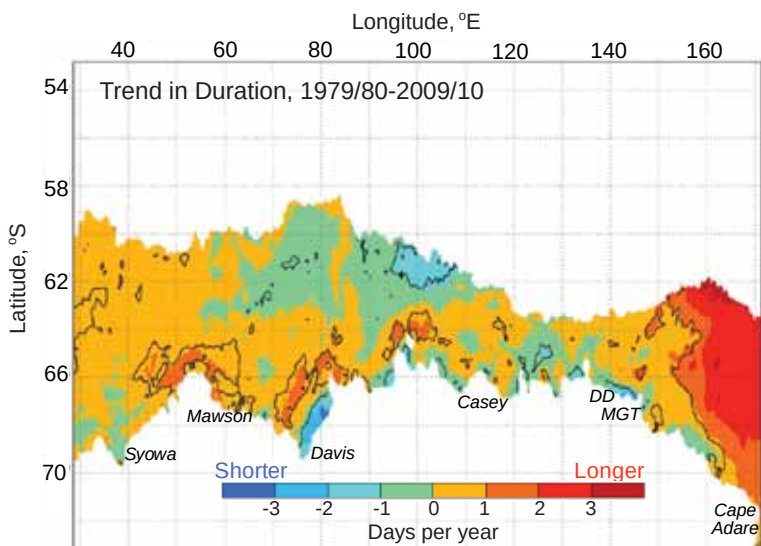


Figure 6. Map of trends in sea ice season duration across East Antarctica, for the period 1979/80–2009/10. Contours denote statistical significance at the 95% level. DD is Dumont d'Urville and MGT Mertz Glacier Tongue. From Massom *et al.* [2013].

While the interplay between large-scale atmospheric patterns (like SAM and ENSO) to some extent explains sea ice variability in the South Pacific sector of the Southern Ocean, it does not explain regional changes observed elsewhere around Antarctica. Moreover, it fails to account for the potentially important role of ocean forcing and feedbacks. The observed magnitude and timing of seasonal/regional sea ice changes in fact point to involvement of strong positive feedbacks. For example, a strong relationship between spring sea ice retreat and subsequent autumn advance observed in the AP-BS (and WRS) regions is consistent with an earlier (later) spring retreat leading to increased (decreased) solar heating of the upper ocean, which then results in a later (earlier) sea ice advance in the following autumn [Stammerjohn *et al.*, 2012].

Sea ice thickness and volume

In the Arctic, a combination of submarine sonar and extensive satellite altimeter data has revealed a substantial thinning of the ice cover there that is consistent with the loss of perennial ice from the Arctic Basin [Rothrock *et al.*, 2008; Kwok and Rothrock, 2009]. However, no such data are available for the Antarctic sea ice zone, where our current knowledge of sea-ice thickness is limited to compilations of various *in-situ* and remote sensing observations [for example Worby *et al.*, 2008] that are characterised by large spatio-temporal gaps and may be biased towards thinner ice [Maksym *et al.*, 2012], and ice chart and satellite altimeter data that are limited to certain sectors or short time periods [for example Zwally *et al.*, 2008; DeLiberty *et al.*, 2011; Yi *et al.*, 2011; Kurtz and Markus, 2012]. As such, important large-scale changes in Antarctic sea-ice thickness and volume may be going unnoticed. Satellite remote sensing, using laser and radar altimeters such as those onboard ICESat and CryoSat 2 to accurately measure surface elevation relative to sea level [Yi *et al.*, 2011; Laxon *et al.*, 2013], clearly represents the only practical means of bridging this critical gap [Figure 7]. Unfortunately, however, there are considerable challenges to deriving accurate and reliable sea-ice thickness estimates from these data in the Antarctic compared to the Arctic. These challenges are assessed later in this document.

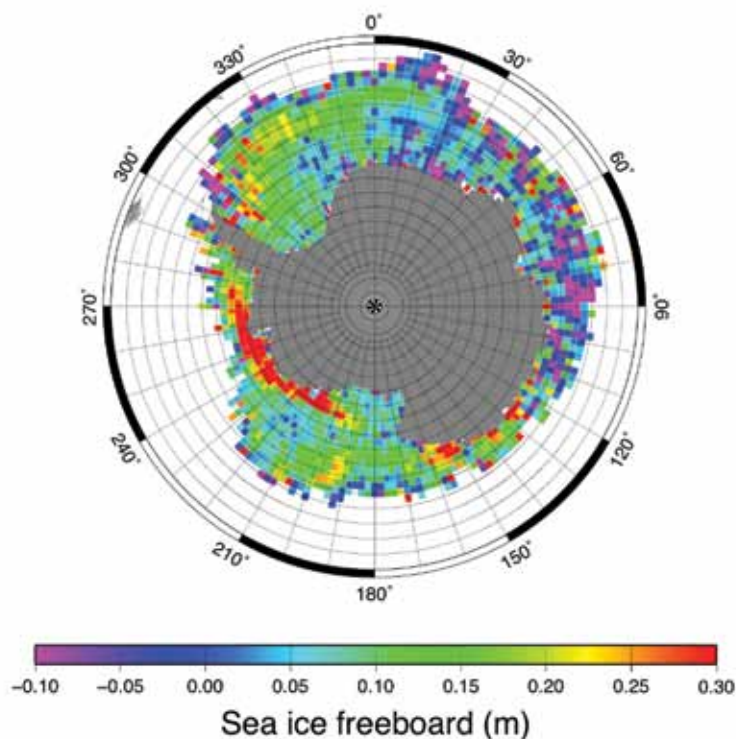


Figure 7. Estimates (uncalibrated) of Antarctic sea ice freeboard (surface height above sea level) at the end of winter in October 2011, derived from CryoSat 2 radar altimeter data. Converting satellite freeboard measurements to ice thicknesses remains a pressing challenge, requiring accurate independent information on snow thickness and volume and ice density [Zwally *et al.*, 2008]. Data such as these only recently became available and present an exciting new development in satellite remote sensing, providing the third dimension of sea ice estimates. Figure courtesy Rachel Tilling, Katharine Giles, Andy Ridout (all CPOM, UCL) and Nathan Kurtz (NASA).

AN INTEGRATED AIRBORNE IMAGING SYSTEM: BRIDGING THE GAP BETWEEN HIGHLY DETAILED IN- SITU OBSERVATIONS AND LARGE SCALE SATELLITE REMOTE SENSING.



The Australian Antarctic program deploys an airborne multi-sensor platform to determine a range of sea ice physical parameters. The instrumented helicopter shown carries a downward-looking scanning LiDAR and thermal infrared temperature sensor (pyrometer) under the cowling in the front, and a downward looking high-resolution, medium format digital camera in the camera bucket below the floor of the helicopter. The LiDAR provides information on combined sea ice and snow cover surface elevation (from which sea ice thickness can be computed); the pyrometer measures the temperature of the surface; and the camera records areal coverage of sea ice and roughness characteristics. A snow thickness radar is a crucial fourth scientific payload and is being developed. This will enable more precise estimates of the snow cover thickness on sea ice, which is one of the important variables when computing sea ice thickness from surface elevation data.

All of these instruments are being used to calibrate and validate satellite remote sensing data, for example from CryoSat-2 (carrying a radar altimeter), or thermal infrared data from polar-orbiting satellites. Such airborne capability is key to linking *in-situ* observations made on a very small local scale (100s of metres) with satellite data on much larger (regional to hemispheric) scales. – to improving geophysical products derived from the latter.

While satellites hold the key to the large-scale monitoring of sea ice thickness, *in-situ* monitoring programmes remain crucially important in the fast-ice zone, where the ice thickness is intimately associated with atmospheric and oceanic circulation and temperature. Under the auspices of the internationally coordinated Antarctic Fast-Ice Network (AFIN) [Heil *et al.*, 2011a], sustained measurements of fast-ice thickness, freeboard and snow thickness are conducted at six locations off the Antarctic coast.

Measurements are taken about once a week. Several AFIN locations indicate increased interannual variability in annual maximum ice thickness over the last 15 years, accompanied by more fast-ice breakout events throughout a year – which reduce the integrated fast-ice volume.

Ice dynamics

In-situ observations of East Antarctic sea-ice drift have shown that atmospheric forcing is the main driver of ice motion [Heil *et al.*, 2009, 2011b], with recent intensification of storm activity in certain regions of East Antarctica leading to increased deformation and ridging (and therefore thickening) of the ice cover [Heil *et al.*, 2011b].

On the broader circum-Antarctic scale, satellite-derived ice motion analysis [Holland and Kwok, 2012] has revealed regional increases in autumn-winter sea ice speed of up to 30% from 1992 to 2009 – most notably in the sectors 20° W–20° E and 90° W–140° E, with decreases of similar magnitude in the western-central part of the Weddell Gyre, the Bellingshausen Sea and off much of East Antarctica (Figure 8). These large and statistically-significant changes are linked to observed changes in surface wind speed and direction associated with change in large-scale modes of climate variability, such as SAM and ENSO [Holland and Kwok, 2012]. Other work carried out within the ACE CRC (Heil *et al.*, in progress) has revealed high interannual variability in ice motion.

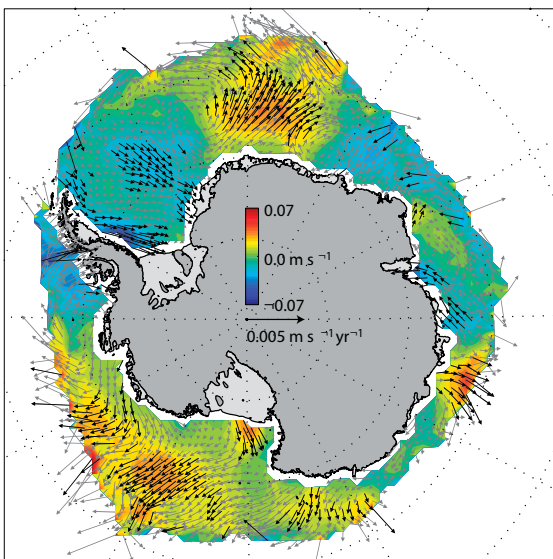


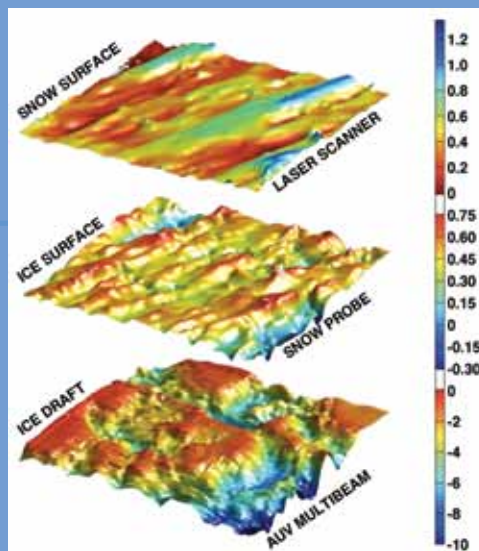
Figure 8. Antarctic autumn-winter (April–October) sea ice motion trend vectors for the period 1992–2010, derived from satellite data and overlain on colour coded 19-year change in meridional ice speed. Figure adapted from Holland and Kwok [2012].

SIPEX 2: A MILESTONE IN UNDERSTANDING ANTARCTIC SEA ICE



The Sea Ice Physics & Ecosystems eXperiment 2 (SIPEX 2) was jointly coordinated by the ACE CRC and the Australian Antarctic Division. This multi-disciplinary study was specifically designed to address major knowledge gaps in Antarctic sea ice zone processes as identified by national and international end-users. It was conducted using RSV *Aurora Australis* in the region of 100° E to 120° E off East Antarctica from September to November 2012, and brought together over 50 scientists from nine nations. The international team measured the physical, biological and biogeochemical properties of sea ice on small-to-regional scales using classical methods and state-of-the-art technology, including ice coring surveys, remotely-operated and autonomous underwater vehicles, drifting buoys and instrumented helicopters. Data from the voyage being processed by ACE CRC and international researchers will enhance our understanding of the role of sea ice in Antarctic climate and ecosystem processes. These data will also be utilised in the validation of satellite-derived sea ice thickness products and the parameterisation of sea ice processes in climate and ecosystem models.

An example of the collaborative and integrative data collected during SIPEX 2 is shown in the graphic below. At the local scale (over a 100 m × 100 m grid), coincident measurement of the ice draft, freeboard and snow loading can provide crucial three-dimensional information at sub-floe scale. This is now possible through the new methodologies for in-situ fieldwork using autonomous underwater vehicles (AUV) and laser and GPS technologies [Williams *et al.*, 2013]. An AUV from the Woods Hole Oceanographic Institution (WHOI), equipped with a swath multi-beam sonar, obtained high-resolution geolocated 3-D maps of Antarctic sea ice draft. Coincident, high-resolution 3-D measurements of snow and ice surface morphology were obtained using terrestrial laser scanners (TLS) and an automated snow probe. The next steps in this approach are to extend the methodology beyond sub-floe scale - to provide regional sampling using, for example, powerful autonomous underwater and airborne instrument packages. The success of these floe-scale missions could pave the way for a new era of field experiments capable of exploring ocean and sea ice processes on scales up to 10–100 kilometres. This increase in scale is important for the goal of directly linking *in-situ* data with satellite sensor footprints and coupled climate model grid cells and ecosystem studies.



Sea ice surface elevation and draft and snow surface elevation data as collected at a typical sea ice station during SIPEX 2 (units in metres). TLS and snow probe data courtesy Ted Maksym (WHOI) and Ernesto Trujillo (CRYOS, EPFL). From Williams *et al.*, 2013.

Fast ice

Recent work conducted at the ACE CRC produced the first high-resolution time series of fast ice maps for the East Antarctic coast (covering the Indian Ocean and Western Pacific Ocean sectors) [Fraser *et al.*, 2012]. This study shows that fast-ice coverage displays large inter-annual variability, but no statistically-significant trend, in the Western Pacific Ocean sector. In contrast, the Indian Ocean sector showed relatively smaller variability, but a statistically-significant positive trend in fast ice extent of $4.1 \pm 0.4\%$ per year. However, these calculations are based on a short period only (~9 years, 2000-2008), and a longer time series is required to conclusively identify long-term trends. Extending the fast ice record both temporally and to circum-Antarctic is an ongoing focus of work at the ACE CRC.

SEA ICE IN SOUTHERN OCEAN BIOGEOCHEMICAL CYCLES AND ECOSYSTEMS

Sea ice is a key driver of Southern Ocean biogeochemical cycles and plays an important role in structuring Antarctic marine ecosystems [Thomas and Dieckmann, 2010]. Antarctic sea ice cover affects the Southern Ocean chemical and biological processes in variable and complex ways [Massom and Stammerjohn, 2010; Vancoppenolle *et al.*, 2013]. Changes in sea ice will therefore have multiple impacts on the Southern Ocean by affecting its elemental cycling, productivity and food web structures. While many coupled sea ice physical-chemical-biological processes have been identified, they are not generally well understood or quantified at large scales. The potential future impacts of changing sea ice on Southern Ocean biogeochemistry and ecosystem structure and function are therefore highly uncertain.

Sea ice biogeochemistry

Recent observations show that iron accumulates in sea ice during autumn and winter and is released back into the ocean in spring [Lannuzel *et al.*, 2007; van der Merwe *et al.*, 2011]. Iron is an essential nutrient for algal growth. The Southern Ocean is a high-nutrient-low-chlorophyll area, which is characterised by high concentrations of macronutrients (for example nitrate and phosphate) but low concentrations of iron. Therefore iron availability is a key factor controlling algal growth in the Southern Ocean and sea ice can act as a major iron store [Lannuzel *et al.*, 2007]. Release of iron during ice melt contributes to the development of ice-edge blooms which serve as ecological hotspots providing food for Antarctic krill (*Euphausia superba*). The krill are in turn exploited by many higher predators including seals, seabirds and whales, which often congregate at retreating ice edges (for example Nicol *et al.* [2008]).

Previously, sea ice was believed to be impermeable to gases. Large-scale ocean-atmosphere carbon dioxide (CO₂) exchange was thought to occur only during the open water season. However, recent studies point towards significant CO₂ fluxes through sea ice, modulated by a permeable sea ice cover and dynamic biological and chemical processes within the sea ice [Delille *et al.*, 2007; Nomura *et al.*, 2010; Rysgaard *et al.*, 2007]. Current knowledge suggests that sea ice is permeable to gas fluxes and can act as both a source of CO₂ in winter and a sink in summer and autumn, and thus affects the

oceanic uptake of this important greenhouse gas in complex ways. On-going research is aimed at determining the processes responsible for the release and uptake of various climate relevant gases from ice-covered oceans [Vancoppenolle *et al.*, 2013].

Sea ice processes are also considered to make a significant contribution to the sulphur cycle through the high production of dimethylsulfoniopropionate (DMSP) by ice algae. DMSP serves as a precursor of sulphate aerosols, which are cloud condensation nuclei and have a potential cooling effect on the planet by increasing cloud albedo [Malin and Kirst, 1997; Stefels *et al.*, 2007; Trevena and Jones, 2006].

Sea ice algae and phytoplankton

Sea ice formation starts in autumn when there are substantial concentrations of micro-organisms in Antarctic surface waters. These organisms are incorporated into the newly forming sea ice by physical processes and create diverse, ice-associated microbial communities (Thomas and Dieckmann, 2010 and references therein). In terms of biomass, these communities are generally dominated by microalgae. In fact, the algae can grow to such high biomasses that they discolour the sea ice (Figure 9).



Figure 9. An overturned first-year sea ice floe showing the high concentration of algae growing on the underside. Photo by Jan L. Lieser (November 2009, East Antarctica).

Observations and model results indicate that 10-20% of the total annual primary production in the ice-covered parts of the Southern Ocean is derived from algae which live within, or are attached to, the bottom of the sea ice [Arrigo and Thomas, 2004; Legendre et al., 1992]. However, these estimates are subject to considerable uncertainty as field observations, which are necessary to validate newly emerging models, remain sparse and are costly to obtain (see Figure 10) [Meiners et al., 2012].

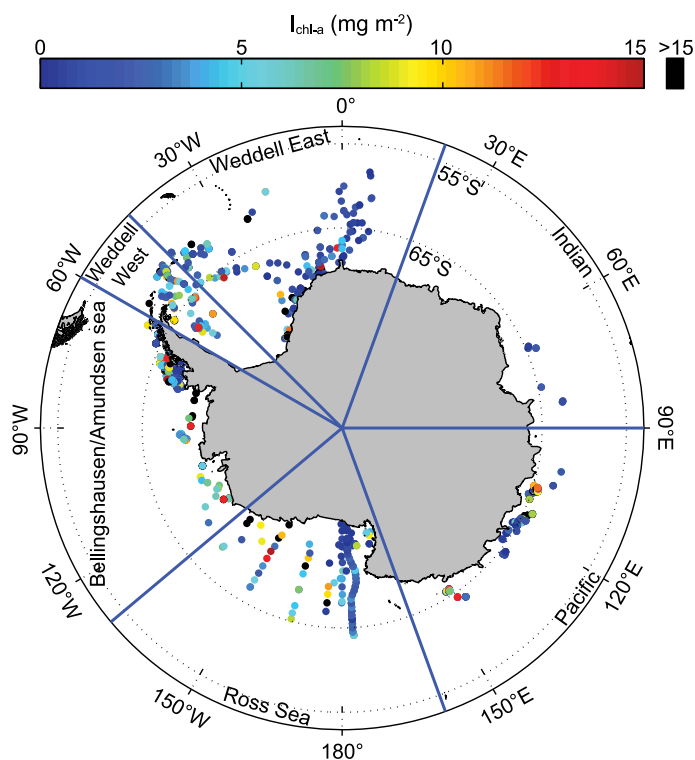


Figure 10. Colour coded dots showing the location of ice cores and the associated ice-algal pigment (ice-algal pigment is measured as milligrams of chlorophyll-a per metre squared). The spatial coverage across six sectors around Antarctica is quite heterogeneous, demonstrating the need for more comprehensive data collection. From Meiners et al. [2012].

While the contribution of sea ice algae to overall primary productivity in the Southern Ocean is considered to be relatively low, these algae provide a critical food source for marine herbivores such as Antarctic krill, especially during winter and early spring when food in the water column is scarce [Flores et al., 2012]. A krill feeding at the underside of an ice floe is shown in Figure 11.

When sea ice melts during spring and summer, low-salinity and less-dense water is added to the ocean surface, which creates stratified conditions that are favourable to algal growth. The melting ice releases nutrients, including iron as well as ice algae, into the stable surface layer of the ocean where the microalgae can continue to grow. In combination, these processes contribute to the formation of major spring-time phytoplankton “blooms” in the vicinity of retreating sea ice edges.



Figure 11. Krill under sea ice. Photo courtesy Hauke Flores (AWI).

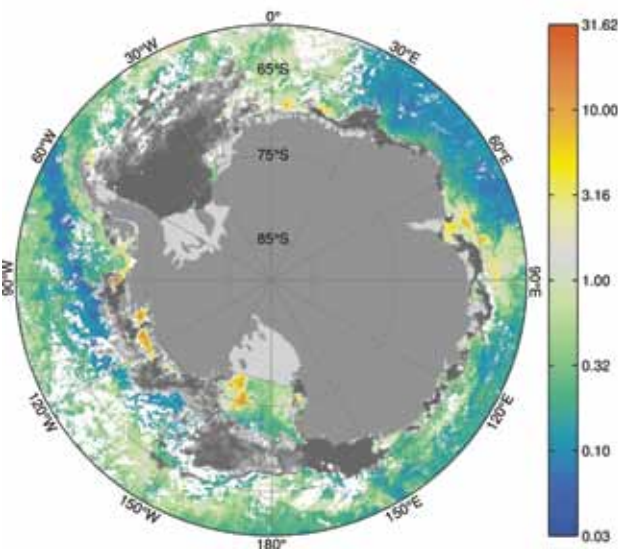


Figure 12. Satellite data composite from January 2009 showing surface ocean colour indicative of algal pigment concentration, which is measured in milligrams of chlorophyll-a per cubic metre. Data are taken from the SeaWiFS satellite project, indicating patchiness in the ice-edge bloom around Antarctica. The sea ice is shown in dark grey. Unfortunately, satellites cannot measure algal biomass in the sea ice zone. Figure courtesy Ben Raymond (AAD).

Ice-edge blooms can provide, at times and in distinct regions, a highly productive environment supporting higher trophic levels and biogeochemical cycling [Arrigo *et al.*, 1998; Smith and Nelson, 1985]. Figure 12 shows the patchiness of chlorophyll-a in a composite of satellite ocean colour data for January 2009, with some hot spots of high productivity for example in the Amundsen Sea, eastern Ross Sea and eastern Davis Sea.

Changes in sea ice extent, thickness and/or duration affect both the timing and magnitude of sea ice algal production and phytoplankton production in the Southern Ocean [Smith and Comiso, 2008; Montes-Hugo *et al.*, 2009]. Models indicate that a decrease in sea ice extent may result in an increase in overall marine primary production due to higher light availability in the ice-free areas [Arrigo and Thomas, 2004; Arrigo *et al.*, 2008]. However, other research indicates that a decrease in sea ice may reduce Southern Ocean iron supply and may reduce overall production [Lancelot *et al.*, 2009; Lannuzel *et al.*, 2010]. Importantly, a decrease in sea ice extent will result in a reduction in the springtime ice-edge algal blooms. While research in the west Antarctic Peninsula region has shown a general decrease in phytoplankton production in relation to decreasing sea ice extent and duration, this response varies significantly along a north-south gradient [Montes-Hugo *et al.*, 2009].

Wildlife and fisheries

Sea ice is a key habitat for Antarctic marine ecosystems, not just in winter but throughout the year [Thomas and Dieckmann, 2010]. Some species such as Crabeater seals (*Lobodon carcinophagus*) breed on the mobile pack ice while others, such as the Weddell seals (*Leptonychotes weddellii*) and Emperor penguins (*Aptenodytes forsteri*) (Figure 13) rely on stable landfast ice as a breeding platform [Kooyman and Burns, 1999]. Sea ice can act as a barrier separating animals from their food source, and can also affect the food source itself. Adélie penguins (*Pygoscelis adeliae*) and Emperor penguins must cross many tens of kilometres of coastal landfast ice from their colonies on the Antarctic continent to reach food supplies for themselves and their newly hatched offspring [Emmerson and Southwell, 2008; Massom *et al.*, 2009]. In fact, a close relationship has been shown to exist between annual fast ice extent/trekking distance and Emperor penguin breeding success at Dumont d'Urville Station [Massom *et al.*, 2009].



Figure 13. Emperor Penguin (*Aptenodytes forsteri*) mating pair with newly hatched chick. Photo by Jan L. Lieser.

A close relationship between winter sea ice extent and the biomass, condition and reproductive success of krill has been demonstrated in the Antarctic Peninsula region and off East Antarctica [Atkinson *et al.*, 2004; Nicol *et al.*, 2000]. Sea ice provides ice algae as a food source for krill and also serves as a refuge from predation for air-breathing vertebrates, which have difficulty accessing the population during the winter [Flores *et al.*, 2012; Nicol, 2006]. Baleen whales migrating south from their tropical wintering grounds aggregate along the edge of the pack ice as it retreats in spring, to feed on krill that have spent their winter under the frozen sea [Nicol *et al.*, 2008]. Changes in sea ice extent are therefore likely to affect ice-associated baleen whale species such as Minke whales (*Balaenoptera bonaerensis*) and Blue whales (*Balaenoptera musculus*), as well as Orca whales (*Orcinus orca*).

Changes in sea ice extent in the west Antarctic Peninsula region have also had consequences for the Antarctic krill fishery. In particular, reduced ice cover has allowed increased access to fishing grounds during winter [Kawaguchi *et al.*, 2009; Nicol *et al.*, 2012]. The ecosystem consequences of changes in fishing activity combined with changes in sea ice habitats are somewhat unclear, however [Melbourne-Thomas *et al.*, 2013; Trivelpiece *et al.*, 2011].

Southern Ocean food webs and the life cycles of many Antarctic marine animals are closely attuned to the seasonal rhythms of their physical environment, in particular the annual advance and retreat of sea ice. The sensitivity of Antarctic ecosystems to changing sea ice conditions has become increasingly apparent in recent years. For example, in the west Antarctic Peninsula region, the trend towards reduced sea ice has resulted in population changes for ice-dependent Adélie penguins. Specifically, there is some evidence that populations of these penguins have moved southwards – to be replaced by Gentoo penguins (*Pygoscelis papua*) and Chinstrap penguins (*Pygoscelis antarctica*) that previously bred predominantly on northerly sub-Antarctic islands [McClintock *et al.*, 2008]. However, the ecosystem consequences of changes in sea ice habitats on the west Antarctic Peninsula region are complicated [Melbourne-Thomas *et al.*, 2013; Montes-Hugo *et al.*, 2009; Trivelpiece *et al.*, 2011], and it is unlikely that observations made in this region can be extrapolated to other regions of the Southern Ocean. Regionally-specific process and modelling studies, and importantly long-term and coordinated observations, are needed to understand and predict the impacts of changing sea ice on Antarctic ecosystem structure and function.

FUTURE IMPACTS OF CLIMATE CHANGE ON ANTARCTIC SEA ICE

Model issues and predictions

Accurately predicting the future state and distribution of Antarctic sea ice represents a considerable challenge for climate modellers. Although sophisticated, the current generation of coupled climate models exhibit a wide range of Antarctic sea ice climatologies, with few reproducing the increase in overall sea ice extent observed since the late 1970s [Eisenman *et al.*, 2011; Maksym *et al.*, 2012; Turner *et al.*, 2013b]. Most climate models fail to accurately simulate mean ice extent, particularly in summer, and

overestimate its year-to-year variability [Maksym *et al.*, 2012]. These factors, which suggest that key processes are being simulated incorrectly and/or neglected, unfortunately undermine the degree of confidence that can be placed in predictions of likely sea-ice conditions over the coming decades [Turner *et al.*, 2013a]. They also underline a critical need to carry out coordinated in-situ measurement programs to better understand sea ice-atmosphere-ocean processes at play.

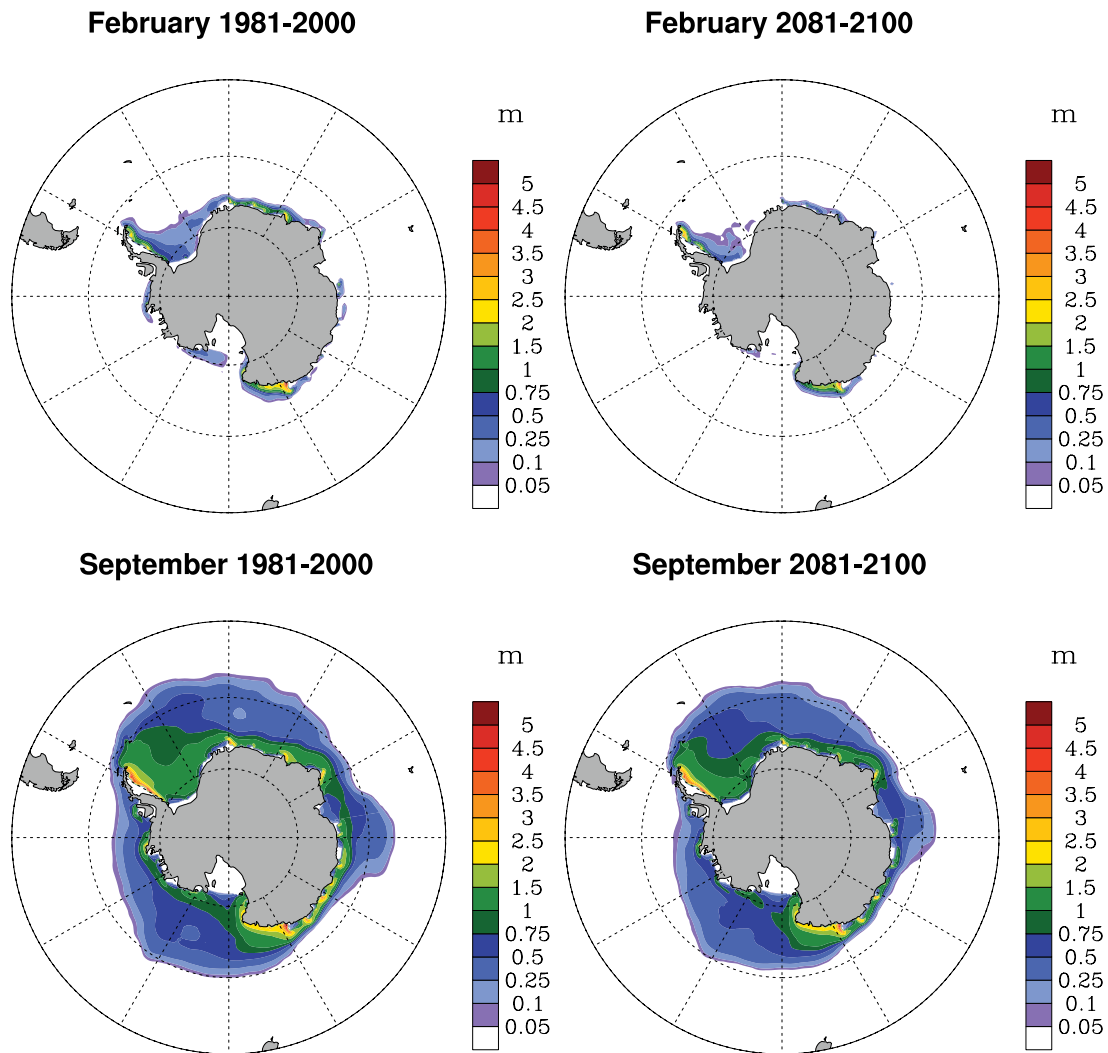


Figure 14. Predicted changes in Antarctic sea ice extent and concentration from the Australian Community Climate and Earth System Simulator. Figure courtesy Siobhan O'Farrell (CSIRO).

Nevertheless, climate models generally predict an overall reduction in Antarctic sea ice by the end of this century, as the effects of greenhouse gas increases become more significant and seasonal influences of stratospheric ozone loss decline (due to recovery of the Antarctic Ozone Hole). Simulations and projections from one model, the Australian Community Climate and Earth-System Simulator or ACCESS model (*Bi et al.*, 2013), are shown in Figure 14. An ensemble mean of the IPCC AR4 models suggest that the annual average total sea ice area will decline by a third by 2100 [*Bracegirdle et al.*, 2008]. This retreat is predicted to be greatest in spring and winter when annual ice extent is largest, which would decrease the amplitude of the seasonal cycle of sea ice areal extent. However, current climate models are unable to provide a precise regional (and seasonal) assessment of expected changes [*Lefebvre and Goosse*, 2008; *Turner et al.*, 2013a]. In terms of seasonality, *Stammerjohn et al.* [2012] speculate that current trends towards later autumn advance and earlier spring retreat may well continue. Future coverage of fast ice is unknown, as it is not included in global climate models.

Model results suggest that ice thickness may be even more sensitive to climate change than ice extent [*Arzel et al.*, 2006]. In their study of ensemble output from IPCC AR4 coupled climate models, *Sen-Gupta et al.* [2009] showed a robust decrease in average ice thickness and volume across the models for both summer and winter. The multi-model mean further indicates that for the winter season, substantial sea ice loss will occur at all longitudes, with the greatest loss occurring in the western Antarctic - particularly from the outer edge of the Weddell region (0° W to 30° W) and between 90° W and 150° W. In summer, the greatest ice loss is predicted to be to the east of the Antarctic Peninsula. As highlighted earlier, accurate assessments of change in Antarctic sea ice thickness and volume are severely hampered by the lack of baseline information [*Maksym et al.*, 2012].

Other likely impacts

Predicted increases in wind strength/storminess in summer and autumn in particular [*Bracegirdle et al.*, 2008; *Turner et al.*, 2013a], waviness and snowfall [*Bracegirdle et al.*, 2008; *Emori and Brown*, 2005; *Meehl et al.*, 2006] are likely to have a major effect on the future Antarctic sea ice environment, and in a complex fashion [*Massom and Stammerjohn*, 2010]. Stronger winds and/or greater prevalence of extreme wind events would lead to more ice rafting and ridging and increased ice thickness in some areas (for example *Massom et al.* [2006]), while greatly affecting the floe size distribution of the sea ice cover. Greater wave energy generated by an increase in storminess would affect the width and characteristics of the marginal ice zone and even the characteristics of the inner pack (for example *Massom et al.*, 1999), as well as the breakup of fast ice (particularly when the latter is unprotected by an extensive pack ice cover, seasonally or otherwise). Fast-ice formation and seasonality are also highly sensitive to wind speed and direction [*Heil*, 2006; *Massom et al.*, 2010]. Ice conditions, especially in bays, will be substantially different in a climate with different wind patterns than at present. The breakup or enhanced melt of fast ice could, however, be counterbalanced by the increased discharge of icebergs since grounded icebergs act as key anchor points for fast ice formation in many locations [*Giles et al.*, 2008]. Increased sheet discharge (potential iceberg production) is predicted to occur under warming conditions [*Bentley et al.*, 2007].

Impact of increased snow

Increased snow accumulation on sea ice could potentially lead to an increase in surface flooding by depressing the ice surface below sea level and warming the ice (a warmer ice cover is more permeable to the passage of seawater through it [Golden *et al.*, 1998]). In this way, increased snowfall could lead to enhanced formation of snow ice [Fichefet and Morales Maqueda, 1999]. This can be a more efficient means of ice thickening, given that freezing occurs on the ice surface rather than the base [Lytle and Ackley, 2001]. In other words, enhanced snow ice formation could maintain sea ice thickness to some (unknown) extent by counterbalancing any increase in ice basal melting under ocean warming conditions [Maksym and Markus, 2008; Wu *et al.*, 1999].

At the same time, the presence of a thicker snow cover could delay the summer ice melt due to its high albedo – to potentially lengthen the sea ice season in certain regions [Eicken *et al.*, 1995; Ledley, 1991]. Another potential impact of greater snow accumulation is an increase in freshwater flux into the ocean when the ice melts, leading to increased ocean surface stratification. This could in turn lead to reductions in the upward flux of ocean heat for basal sea ice melt, implying that sea ice duration and extent may actually increase under warming conditions (for example Zhang [2007]). These potential scenarios once again underline the complexity of the sea ice-ocean-atmosphere interaction system.

FUTURE CHALLENGES

Improving the performance of models

In spite of major advances in our understanding of Antarctic sea ice over the past three decades, there is still much to be learned about its characteristics; its interactions and connections with the atmosphere, ocean and ecosystem; and its complex response to, and role in, present and future climate change [Maksym *et al.*, 2012]. A key challenge is to improve the performance of numerical models in simulating recent, and predicting future, Antarctic sea ice distribution and state, and to include seasonal and regional sensitivities in the models [Maksym *et al.*, 2012; Turner *et al.*, 2013a]. Improved parameterisation and understanding of change and variability not only requires more and improved observations. It also necessitates identification and better understanding of processes and mechanisms involved. These include ocean forcing and feedbacks [Russell *et al.*, 2006], as well as the role that sea ice itself may play in modulating the response to external (atmospheric and oceanic) forcing [Raphael *et al.*, 2011; Maksym *et al.*, 2012].

Estimating sea ice thickness

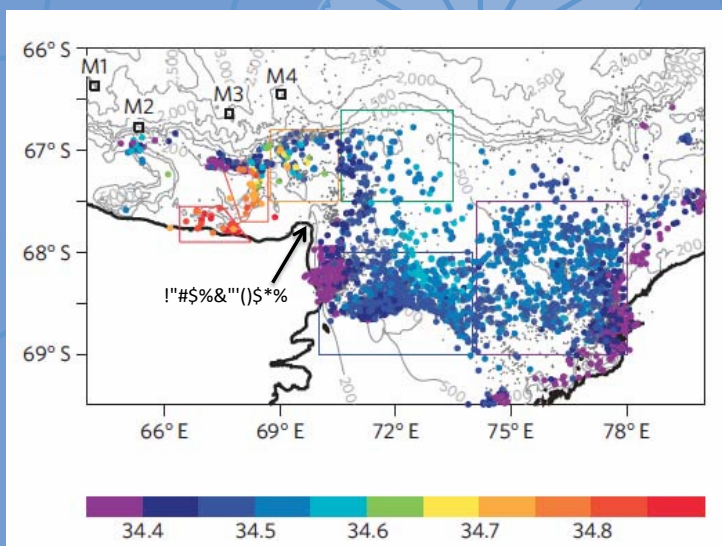
A major stumbling block at present is our lack of knowledge of the current distribution of the thickness of Antarctic sea ice and its snow cover, and of how these are changing and what processes are involved; that is, the relative contributions of dynamic and thermodynamic processes (including snow-ice formation). The ice thickness distribution provides an integrated measure of total sea ice production and, thus, total ocean-atmosphere heat loss, the surface salinity flux in winter and freshwater input into the ocean

SEALS, POLYNYS AND A 30-YEAR-OLD MYSTERY OF THE DEEP

Scientists believed that the cold, dense bottom waters of the global ocean originated at three different locations in Antarctica - the Weddell Sea, the Ross Sea and the Adélie Coast of East Antarctica. Thirty years ago, a fourth source was speculated to exist somewhere in the Prydz Bay region, but until now scientists have been unable to confirm if, where and how it is being formed.

Now, through sophisticated satellite data, oceanographic moorings and tagged seals, a team involving ACE CRC researchers has discovered that a fourth stream of "Antarctic Bottom Water" (AABW) is being produced from intense sea ice formation in the Cape Darnley Polynya (65° E–69° E, 67.5° S), north-west of the Amery Ice Shelf. Oshima et al. [2013] estimate that this source of AABW represents between 6% and 13% of the circumpolar total. Its significance is underlined by the fact that AABW is a key driver of the global ocean circulation and therefore of the Earth's climate.

Data collected by instrumented Southern elephant seals allowed scientists to pinpoint the path of the dense salty water and therefore identify the area from which it originates. This oceanographic data collected by the seals is part of a larger ecological project at Institute for Marine and Antarctic Studies (IMAS) to study their behaviour. When the seals surface, their instrument tags relay information (via satellite) back to land, where it is collected by the Integrated Marine Observing System (IMOS). The newly identified source of AABW is different from the other three sources. Its existence demonstrates that polynyas are capable of forming sufficiently dense Shelf Water over a narrow section of continental shelf without the traditional assistance of a large ice shelf or coastal storage volume. This opens the possibility for further discoveries of AABW production from the other polynya regions around the Antarctic coastline.



Salinities at bottom-of-dive for all available ocean temperature and salinity profiles measured by instrumented seals over the continental shelf off Cape Darnley and in Prydz Bay where the bottom depth was >250m and potential temperature <1.7 °C. All other dive locations are indicated by small grey points. From Oshima et al. [2013].

in summer. Current large observational uncertainties in these quantities (of 50%-100%) severely compromise the ability to effectively evaluate models - and thus undermine our confidence in the accuracy of future predictions [Maksym *et al.*, 2012].

Given these factors, it is critical to derive more accurate baseline information on the thickness and volume of sea ice on regional to circumpolar scales by satellite remote sensing. This can only be achieved through effective calibration and validation programs involving careful coordination of *in-situ* observations with coincident satellite, airborne and under-ice remote sensing (for example Lieser *et al.* [2011, 2013], Williams *et al.* [2013]). This in itself represents a considerable challenge, given the range of scales involved. Other key challenges to the derivation of sea ice thickness information from satellite altimetry (both laser and radar) are the relatively small freeboard of Antarctic (compared to Arctic) sea ice, the widespread (though as yet unquantified) occurrence of surface flooding, and the need for accurate independent information on snow thickness and density and ice density [Giles *et al.*, 2008a; Maksym and Markus, 2008].

Other challenges

Clearly, improved knowledge of the precise mechanisms involving, and feedbacks between, snowfall, ice growth and melt, and upper-ocean stability (stratification) and vertical heat transport is critical for understanding the current behaviour and characteristics of Antarctic sea ice and its future trajectory [Maksym *et al.*, 2012]. This again represents a considerable challenge, given the complexities and regional and seasonal sensitivities involved. Fortunately, we are on the cusp of an exciting new era in observational capability for monitoring the Antarctic sea ice environment, using autonomous drifting and ice-tethered platforms [Maksym *et al.*, 2012]. These can provide continuous, season-long observations of atmosphere-ice-ocean interactions and the evolution of the sea ice and its snow cover that have been largely lacking to date. While such platforms have the potential to revolutionise our understanding of the severely under-sampled Antarctic sea ice zone, relatively few have been deployed to date. The challenge is again to step up deployment, but in a coordinated fashion around Antarctica (including on fast ice and in the marginal ice zone).

Given the rapid decline in Arctic sea ice and our uncertain knowledge of Antarctic sea ice, there is strong impetus for process studies and more sustained integrated observations that span disciplines and a range of space and time scales. The continuation of field-based observations using icebreakers is crucial in this respect [Maksym *et al.*, 2012].

STRATEGIC FOCUS

The Intergovernmental Panel on Climate Change concluded in its Fifth Assessment Report [IPCC, 2013] that the observed warming of the Earth is strongest in the polar regions. Surface temperatures over large areas of the Antarctic Peninsula have risen considerably faster than the global average. The Southern Ocean is also warming more rapidly than the global ocean average.

Antarctic and Southern Ocean processes influence weather and climate on both regional and global scales, and impact the wider environment and ecosystems.

The Antarctic sea-ice zone is a major component of the Earth's climate system, and we know that it is changing. As this Position Analysis shows, there are key components of Antarctic sea ice that we know little about. Filling this lack of knowledge of such a vital region of the globe is an important research task for the coming decade.

In recent years, the technology needed to observe the ocean has developed rapidly. The use of this new technology has the potential to reduce the cost of research in the region. Platforms ranging from drifting buoys, to autonomous underwater robots and remotely-operated aircraft can be used to complement traditional (and essential) icebreaker research platforms and aircraft. These on-site fieldwork measurements can be matched with satellite technologies to observe the sea-ice zone on the hemispheric and global scale.

It will be essential to include many different disciplines in this effort, as feedbacks across various regional scales will link into the different spheres of the global system. This research must be integrated with national and international efforts in the region. While satellites can measure sea-ice extent, it will be important to get a measure of and develop methods for predicting sea ice thickness in the entire sea-ice zone of the Antarctic. This new knowledge will support improvements to numerical models of the coupled atmosphere-ocean-cryosphere system, and through this, the predictability of the climate system and, as an extension, also that of ecosystem models.

Australian research is important in understanding impacts arising from changes in Antarctic sea ice, but Australia also has a key role in ensuring that such research is available in relevant forums of the Antarctic Treaty System (ATS), most notably the Antarctic Treaty Consultative Meeting and the Commission for the Conservation of Antarctic Marine Living Resources (CCAMLR).

Earlier work by the ACE CRC noted that changes in sea ice are likely to impact on a number of activities within the Antarctic Treaty area. Reductions in sea ice will reduce 'natural' barriers to shipping access in high latitudes and open up otherwise difficult areas for marine resources harvesting. While these activities are managed under the ATS, Australia has an interest in ensuring that these decisions take account of current science. Australia's commitment to integrated research, for example the two SIPEX voyages (in 2007 and 2012), provides crucially important contributions to international efforts to understand observed and potential changes to sea ice. This research and its outcomes are directly linked to Australian interests and government goals for the Australian Antarctic Program.

Australia's work on sea ice builds on a lengthy commitment to marine science and Antarctic infrastructure. While such commitments are challenging, Australia will be well served by maintaining capacity and assets in high-latitude marine science with sufficient

capacity to undertake winter sea ice expeditions. This means maintaining sufficient icebreaking capacity in its ongoing Antarctic logistics and infrastructure planning, and embracing the use of new technologies that greatly aid measurement and monitoring of the difficult sea-ice environment.

Such investments provide opportunities to work with international partners in challenging environments and to continue a century of scientific endeavour that addresses critical questions and supports management of a region that is of direct interest to Australia.

CONCLUSION

Antarctic sea ice is regarded as a sensitive indicator of climate change. Its presence at the interface between the atmosphere and the ocean in polar latitudes makes it highly susceptible to dynamic and thermodynamic changes and variations from above and below. It responds to changes in the environmental conditions in a complex way. For example, while a warming atmosphere and increased heat content in the ocean will have a thinning effect on the sea ice, greater wind and wave energy can break up sea ice to potentially increase the ice thickness by rafting and ridging.

An increase in snow cover on top of the sea ice will help to thermally insulate the ocean from the atmosphere and therefore reduce freezing rates of sea ice, but at the same time can suppress a thinner ice cover below the water line and facilitate increase in ice thickness by snow-ice formation.

The observed patterns of change in Antarctic sea ice are manifold and not uniform across the region. While the overall extent of sea ice, for example, shows a slight increasing trend (panel a. in Figure 4), there are starkly contrasting trends on oceanic basin scales (Ross Sea versus Bellingshausen/Amundsen seas). Similar heterogeneous patterns can be found in maps of trends in sea ice duration/seasonality (Figures 5 and 6).

Sea ice plays a critical role in biogeochemical cycles and the marine ecosystem in the high-latitude Southern Ocean. The life cycles of many marine animals are closely linked to the seasonal rhythm of the physical environment, and changing sea ice conditions might have beneficial effects for some species, but can be potentially devastating for others.

Recent advances in space-borne remote sensing enable a first assessment of large-scale sea ice thickness (and subsequently sea ice volume) based on surface elevation (freeboard) measurements. Such measurements will enable a reliable estimate of sea ice volume over time if carried out in the same area repeatedly with constant precision.

In order to better understand the influence of a changing environment on the geophysical properties of sea ice and flow-on effects on the ice-associated ecosystem, as well as to calibrate and validate established and emerging remote-sensing products, a continued internationally collaborative and inter-disciplinary *in-situ* monitoring and research effort is crucial.



Scientists and support staff at work on East Antarctic sea ice during the second Sea Ice Physics & Ecosystems eXperiment (SIPEX 2) in 2012. Photo by Brian Walpole

REFERENCES

- Abram, N. J., E. R. Thomas, J. R. McConnell, R. Mulvaney, T. J. Bracegirdle, L. C. Sime, and A. J. Aristarain (2010), Ice core evidence for a 20th Century decline of sea ice in the Bellingshausen Sea, Antarctica, *Journal of Geophysical Research*, *115* (D23), D23101, doi:10.1029/2010JD014644.
- Allison, I., R. E. Brandt, and S. G. Warren (1993), East Antarctic sea ice: Albedo, thickness distribution, and snow cover, *Journal of Geophysical Research*, *98*(C7), 12417–12429.
- Arrigo, K. R., and D. N. Thomas (2004), Large scale importance of sea ice biology in the Southern Ocean, *Antarctic Science*, *16*(4), 471–486, doi:10.1017/S0954102004002263.
- Arrigo, K. R., G. L. van Dijken, and S. Bushinsky (2008), Primary production in the Southern Ocean, 1997 – 2006, *Journal of Geophysical Research*, *113*, 1–27, doi:10.1029/2007JC004551.
- Arrigo, K. R., A. M. Weiss, and W. O. Smith Jr. (1998), Physical forcing of phytoplankton dynamics in the southwestern Ross Sea, *Journal of Geophysical Research*, *103*(C1), 1007–1021.
- Arzel, O., T. Fichefet, and H. Goosse (2006), Sea ice evolution over the 20th and 21st centuries as simulated by current AOGCMs, *Ocean Modelling*, *12*, 401–415, doi:10.1016/j.ocemod.2005.08.002.
- Atkinson, A., V. Siegel, E. Pakhamov, and P. Rothery (2004), Long-term decline in krill stock and increase in salps within the Southern Ocean, *Nature*, *432*, doi:10.1038/nature02950.1.
- Bentley, C. R., R. H. Thomas, and I. Veliconga (2007), Ice Sheets, in *Global Outlook for Ice and Snow*, pp. 100–113.. United Nations Environment Programme.
- Bi, D., M. Dix, S. Marsland, S. O'Farrell, H. Rashid, P. Uotila, A. Hirst, E. Kowalczyk, M. Golebiewski, A. Sullivan, H. Yan, N. Hannah, C. Franklin, Z. Sun, P. Vohralik, I. Watterson, X. Zhou, R. Fiedler, M. Collier, Y. Ma, J. Noonan, L. Stevens, P. Uhe, H. Zhu, S. Griffies, R. Hill, C. Harris and K. Puri (2013), The ACCESS coupled model: description, control climate and evaluation, *Australian Meteorological and Oceanographic Journal*, *63*, 41–64.
- Bintanja, R., G. J. van Oldenborgh, S. S. Drijfhout, B. Wouters, and C. A. Katsman (2013), Important role for ocean warming and increased ice-shelf melt in Antarctic sea-ice expansion, *Nature Geoscience*, *6*(4), 1–4, doi:10.1038/ngeo1767.
- Bracegirdle, T. J., W. M. Connolley, and J. Turner (2008), Antarctic climate change over the twenty first century, *Journal of Geophysical Research*, *113*(D3), D03103, doi:10.1029/2007JD008933.
- Cavalieri, D. J., and C. L. Parkinson (2008), Antarctic sea ice variability and trends, 1979 – 2006, *Journal of Geophysical Research*, *113*, 1–19, doi:10.1029/2007JC004564.
- Comiso, J. C. (2010), Variability and trends of the global sea ice cover, in *Sea Ice*, edited by D. N. Thomas and G. S. Dieckmann, pp. 205–246, Wiley-Blackwell, Oxford.
- Comiso, J. C., and F. Nishio (2008), Trends in the sea ice cover using enhanced and compatible AMSR-E, SSM/I, and SMMR Data, *Journal of Geophysical Research* *113*:1–22. doi:10.1029/2007JC004257.
- Comiso, J. C., R. Kwok, S. Martin, and A. L. Gordon (2011), Variability and trends in sea ice extent and ice production in the Ross Sea, *Journal of Geophysical Research*, *116*(C4), 1–19, doi:10.1029/2010JC006391.
- Cook, A. J., and D. G. Vaughan (2010), Overview of areal changes of the ice shelves on the Antarctic Peninsula over the past 50 years, *The Cryosphere*, *4*(1), 77–98, doi:10.5194/tc-4-77-2010.
- Curran, M. A. J., T. D. van Ommen, V. I. Morgan, K. L. Phillips, and A. S. Palmer (2003), Ice core evidence for Antarctic sea ice decline since the 1950s., *Science*, *302*(5648), 1203–6, doi:10.1126/science.1087888.
- de la Mare, W. K. (1997), Abrupt mid-twentieth-century decline in Antarctic sea-ice extent from

- whaling records, *Nature*, 389(September), 57–60.
- de la Mare, W. K. (2008), Changes in Antarctic sea-ice extent from direct historical observations and whaling records, *Climatic Change*, 92(3-4), 461–493, doi:10.1007/s10584-008-9473-2.
- DeLiberty, T. L., C. A. Geiger, S. F. Ackley, A. P. Worby, and M. L. Van Woert (2011), Estimating the annual cycle of sea-ice thickness and volume in the Ross Sea, *Deep Sea Research Part II: Topical Studies in Oceanography*, 58(9-10), 1250–1260, doi:10.1016/j.dsr2.2010.12.005
- Delille, B., B. Jourdain, A. V. Borges, J.-L. Tison, and D. Delille (2007), Biogas (CO₂, O₂, dimethylsulfide) dynamics in spring Antarctic fast ice, *Limnology and Oceanography*, 52(4), 1367–1379, doi:10.4319/llo.2007.52.4.1367.
- Eicken, H., H. Fischer, and P. Lemke (1995), Effects of the snow cover on Antarctic sea ice and potential modulation of its response to climate change, *Annals of Glaciology*, 21, 369–376.
- Eisenman, I., T. Schneider, D. S. Battisti, and C. M. Bitz (2011), Consistent changes in the sea ice seasonal cycle in response to global warming, *Journal of Climate*, 24(20), 5325–5335, doi:10.1175/2011JCLI4051.1.
- Emmerson, L., and C. Southwell (2008), Sea ice cover and its influence on Adélie penguin reproductive performance, *Ecology*, 89(8), 2096–2102.
- Emori, S., and S. J. Brown (2005), Dynamic and thermodynamic changes in mean and extreme precipitation under changed climate, *Geophysical Research Letters*, 32(17), L17706, doi:10.1029/2005GL023272.
- Fichefet, T., and M. A. Morales Maqueda (1999), Modelling the influence of snow accumulation and snow-ice formation on the seasonal cycle of the Antarctic sea-ice cover, *Climate Dynamics*, 15(4), 251–268, doi:10.1007/s003820050280.
- Flores, H., J. A. van Franeker, V. Siegel, M. Haraldsson, V. Strass, E. H. Meesters, U. Bathmann, and W. J. Wolff (2012), The association of Antarctic krill *Euphausia superba* with the under-ice habitat, *PLoS ONE*, 7(2), e31775, doi:10.1371/journal.pone.0031775.
- Fraser, A. D., R. A. Massom, K. J. Michael, B. K. Galton-Fenzi, and J. L. Lieser (2012), East Antarctic landfast sea ice distribution and variability, 2000–08, *Journal of Climate*, 25(4), 1137–1156, doi:10.1175/JCLI-D-10-05032.1.
- Galton-Fenzi, B.K., J. R. Hunter, R. Coleman, S. J. Marsland and R. Warner (2012), Numerical modelling of melt/freeze beneath the Amery Ice Shelf. *Journal of Geophysical Research - Oceans*, 117, C09031, doi:10.1029/2012JC008214.
- Giles, A. B., R. A. Massom, and V. I. Lytle (2008), Fast-ice distribution in East Antarctica during 1997 and 1999 determined using RADARSAT data, *Journal of Geophysical Research*, 113, 1–15, doi:10.1029/2007JC004139.
- Giles, K. A., S. W. Laxon, and A. P. Worby (2008a), Antarctic sea ice elevation from satellite radar altimetry, *Geophysical Research Letters*, 35, pp 1-5.
- Gillett, N. P., J. F. Scinocca, D. A. Plummer, and M. C. Reader (2009), Sensitivity of climate to dynamically-consistent zonal asymmetries in ozone, *Geophysical Research Letters*, 36(10), L10809, doi:10.1029/2009GL037246.
- Golden, K. M., S. F. Ackley, and V. I. Lytle (1998), The percolation phase transition in sea ice, *Science*, 282(5397), 2238–41.
- Heil, P. (2006), Atmospheric conditions and fast ice at Davis, East Antarctica: A case study, *Journal of Geophysical Research*, 111(C5), C05009, doi:10.1029/2005JC002904.
- Heil, P., and I. Allison (1999), The pattern of Antarctic sea-ice drift in the Indian Ocean and western

- Pacific sectors, *Journal of Geophysical Research*, 104(C7), 15789–15802.
- Heil, P., R. A. Massom, I. Allison, A. P. Worby, and V. I. Lytle (2009), Role of off-shelf to on-shelf transitions for East Antarctic sea ice dynamics during spring 2003, *Journal of Geophysical Research*, 114(C9), C09010, doi:10.1029/2008JC004873.
- Heil, P., S. Gerland, and M. A. Granskog (2011a), An Antarctic monitoring initiative for fast ice and comparison with the Arctic, *The Cryosphere Discussions*, 5(5), 2437–2463, doi:10.5194/tcd-5-2437-2011.
- Heil, P., R. A. Massom, I. Allison, and A. P. Worby (2011b), Physical attributes of sea-ice kinematics during spring 2007 off East Antarctica, *Deep Sea Research Part II: Topical Studies in Oceanography*, 58(9-10), 1158–1171, doi:10.1016/j.dsr2.2010.12.004.
- Holland, P. R., and R. Kwok (2012), Wind-driven trends in Antarctic sea-ice drift, *Nature Geoscience*, 5(12), 872–875, doi:10.1038/ngeo1627.
- IPCC (2013), *Climate Change 2013: The Physical Science Basis*. Contribution of Working Group 1 to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change, p. 2216, available at <http://www.climatechange2013.org/report>
- Kawaguchi, S., S. Nicol, and A. J. Press (2009), Direct effects of climate change on the Antarctic krill fishery, *Fisheries Management and Ecology*, 16(5), 424–427, doi:10.1111/j.1365-2400.2009.00686.x.
- Kimura, N. (2004), Sea Ice Motion in Response to Surface Wind and Ocean Current in the Southern Ocean, *Journal of the Meteorological Society of Japan*, 82(4), 1223–1231, doi:10.2151/jmsj.2004.1223.
- Kooyman, G. L., and J. Burns (1999), Weddell Sea Versus Emperor Penguin: Boss of the Ross Sea, *American Zoologist*, 39 (9).
- Korhonen, H., K. S. Carslaw, P. M. Forster, S. Mikkonen, N. D. Gordon, and H. Kokkola (2010), Aerosol climate feedback due to decadal increases in Southern Hemisphere wind speeds, *Geophysical Research Letters*, 37(2), 1–6, doi:10.1029/2009GL041320.
- Kurtz, N. T., and T. Markus (2012), Satellite observations of Antarctic sea ice thickness and volume, *Journal of Geophysical Research*, 117(July), 1–9, doi:10.1029/2012JC008141.
- Kwok, R., and D. A. Rothrock (2009), Decline in Arctic sea ice thickness from submarine and ICESat records: 1958 – 2008, *Geophysical Research Letters*, 36, 1–5, doi:10.1029/2009GL039035.
- Lancelot, C., A. de Montety, H. Goosse, S. Becquevort, V. Schoemann, B. Pasquer, and M. Vancoppenolle (2009), Spatial distribution of the iron supply to phytoplankton in the Southern Ocean: a model study, *Biogeosciences*, 6(12), 2861–2878, doi:10.5194/bg-6-2861-2009.
- Lannuzel, D., V. Schoemann, J. de Jong, B. Pasquer, P. van der Merwe, F. Masson, J.-L. Tison, and A. Bowie (2010), Distribution of dissolved iron in Antarctic sea ice: Spatial, seasonal, and inter-annual variability, *Journal of Geophysical Research*, 115(G3), G03022, doi:10.1029/2009JG001031.
- Lannuzel, D., V. Schoemann, J. de Jong, J.-L. Tison, and L. Chou (2007), Distribution and biogeochemical behaviour of iron in the East Antarctic sea ice, *Marine Chemistry*, 106(1-2), 18–32, doi:10.1016/j.marchem.2006.06.010.
- Laxon, S. W., K. A. Giles, A. L. Ridout, D. J. Wingham, R. Willatt, R. Cullen, R. Kwok, A. Schweiger, J. Zhang, C. Haas, S. Hendricks, R. Krishfield, N. Kurtz, S. Farrell, and M. Davidson (2013), CryoSat-2 estimates of Arctic sea ice thickness and volume, *Geophysical Research Letters*, 40, 732–737, doi:10.1002/grl.50193.
- Ledley, T. S. (1991), Snow on sea ice: Competing effects in shaping climate, *Journal of Geophysical Research*, 96(D9), 17195–17208, doi:10.1029/91JD01439.
- Lefebvre, W., and H. Goosse (2008), An analysis of the atmospheric processes driving the large-scale winter sea ice variability in the Southern Ocean, *Journal of Geophysical Research*, 113(C2), 1–15, doi:10.1029/2006JC004032.

- Legendre, L., S. F. Ackley, G. S. Dieckmann, B. Gulliksen, R. Horner, T. Hoshiai, I. A. Melnikov, W. S. Reeburgh, M. Spindler, and C. W. Sullivan (1992), Ecology of sea ice biota, *Polar Biology*, 12, 429–444.
- Lieser, J. L., A. D. Steer, A. P. Worby, and R. A. Massom (2011), Australian CryoSat-2 CAL/VAL efforts in the East Antarctic sea ice zone, in *CryoSat Validation Workshop 2011*, p. 5, ESA SP-693, Frascati, Italy.
- Lieser, J. L., P.W. Jansen, A.D. Steer, P. Heil, R.A. Massom, G.D. Williams and C. Kunz (2013), First results from a CryoSat-2 calibration experiment in the East Antarctic sea ice zone, SIPEX-2, in *CryoSat-2 Third User Workshop 2013*, ESA SP-717, Dresden, Germany.
- Liu, J., and J. A. Curry (2010), Accelerated warming of the Southern Ocean and its impacts on the hydrological cycle and sea ice, *PNAS*, 107(34), 14987–14992, doi:10.1073/pnas.1003336107.
- Liu, J., J. A. Curry, and D. G. Martinson (2004), Interpretation of recent Antarctic sea ice variability, *Geophysical Research Letters*, 31(2), L02205, doi:10.1029/2003GL018732.
- Lytle, V., and S. Ackley (2001), Snow-ice growth: a fresh-water flux inhibiting deep convection in the Weddell Sea, Antarctica, *Annals of Glaciology*, 33, 45–50.
- Maksym, T., and T. Markus (2008), Antarctic sea ice thickness and snow to ice conversion from atmospheric reanalysis and passive microwave snow depth, *Journal of Geophysical Research*, 113, 1–18.
- Maksym, T., S. E. Stammerjohn, S. Ackley, and R. Massom (2012), Antarctic sea ice - A polar opposite?, *Oceanography*, 25(3).
- Malin, G., and G. O. Kirst (1997), Algal production of dimethyl sulfide and its atmospheric role, *Journal of Phycology*, 33, 889–896.
- Marshall, G. J., P. A. Stott, J. Turner, W. M. Connolley, J. C. King, and T. A. Lachlan-Cope (2004), Causes of exceptional atmospheric circulation changes in the Southern Hemisphere, *Geophysical Research Letters*, 31(14), L14205, doi:10.1029/2004GL019952.
- Martinson, D. G. (2012), Antarctic circumpolar current's role in the Antarctic ice system: An overview, *Palaeogeography, Palaeoclimatology, Palaeoecology*, 335–336, 71–74, doi:10.1016/j.palaeo.2011.04.007.
- Massom, R. A., and S. E. Stammerjohn (2010), Antarctic sea ice change and variability - Physical and ecological implications, *Polar Science*, 4(2), 149–186, doi:10.1016/j.polar.2010.05.001.
- Massom, R. A., P. A. Reid, S. E. Stammerjohn, B. Raymond, A. D. Fraser, and S. Ushio (2013), Change and variability in East Antarctic sea ice seasonality, 1979/80 – 2009/10, *PLoS ONE*, 8(5), doi:10.1371/journal.pone.0064756.
- Massom, R. A., A. B. Giles, H. A. Fricker, R. C. Warner, B. Legrésy, G. Hyland, N. Young, and A. D. Fraser (2010), Examining the interaction between multi-year landfast sea ice and the Mertz Glacier Tongue, East Antarctica: Another factor in ice sheet stability?, *Journal of Geophysical Research*, 115(C12), 1–15, doi:10.1029/2009JC006083.
- Massom, R. A., K. Hill, C. Barbraud, N. Adams, A. Ancel, L. Emmerson, and M. J. Pook (2009), Fast ice distribution in Adélie Land, East Antarctica: interannual variability and implications for emperor penguins *Aptenodytes forsteri*, *Marine Ecology Progress Series*, 374, 243–257, doi:10.3354/meps07734.
- Massom, R. A., S. E. Stammerjohn, R. C. Smith, M. J. Pook, R. A. Iannuzzi, N. Adams, D. G. Martinson, M. Vernet, W. R. Fraser, L. B. Quetin, R. M. Ross, Y. Massom, and H. R. Krouse (2006), Extreme anomalous atmospheric circulation in the West Antarctic Peninsula Region in austral spring and summer 2001/02, and its profound impact on sea ice and biota, *Journal of Climate*, 19, 3544–3571.
- Massom, R. A., J. C. Comiso, A. P. Worby, V. I. Lytle, and L. Stock (1999), Regional classes of sea ice

- cover in the East Antarctic pack observed from satellite and in-situ data during a winter time period, *Remote Sensing of Environment*, 68(98), 61–76.
- McClintock, J., H. Ducklow, and W. Fraser (2008), Ecological responses to climate change on the Antarctic Peninsula, *American Scientist*, 96, 302–310.
- Meehl, G. A., W. M. Washington, B. D. Santer, W. D. Collins, J. M. Arblaster, A. Hu, D. M. Lawrence, H. Teng, L. E. Buja, and W. G. Strand (2006), Climate change projections for the twenty-first century and climate change commitment in the CCSM3, *Journal of Climate*, 19(11), 2597–2616, doi:10.1175/JCLI3746.1.
- Meiners, K. M., M. Vancoppenolle, S. Thanassekos, G. S. Dieckmann, D. N. Thomas, J.-L. Tison, K. R. Arrigo, D. L. Garrison, A. McMinn, D. Lannuzel, P. van der Merwe, K. M. Swadling, W. O. Smith, I. Melnikov, and B. Raymond (2012), Chlorophyll a in Antarctic sea ice from historical ice core data, *Geophysical Research Letters*, 39(21), 1–5, doi:10.1029/2012GL053478.
- Melbourne-Thomas, J., A. Constable, S. Wotherspoon, and B. Raymond (2013), Testing paradigms of ecosystem change under climate warming in Antarctica., *PLoS ONE*, 8(2), e55093, doi:10.1371/journal.pone.0055093.
- Meredith, M. P., and J. C. King (2005), Rapid climate change in the ocean west of the Antarctic Peninsula during the second half of the 20th century, *Geophysical Research Letters*, 32(19), L19604, doi:10.1029/2005GL024042.
- Montes-Hugo, M., S. C. Doney, H. W. Ducklow, W. Fraser, D. Martinson, S. E. Stammerjohn, and O. Schofield (2009), Recent changes in phytoplankton Western Antarctic Peninsula, *Science*, 323, 1470–1473.
- Murphy, E. J., A. Clarke, C. Symon, J. Priddle (1995), Temporal variation in Antarctic sea-ice: analysis of a long term fast-ice record from the South Orkney Islands, *Deep-Sea Research Part I: Oceanographic Research Papers*, 42(7)1045-1062, doi:10.1016/0967-0637(95)00057-D.
- Nicol, S. (2006), Krill, currents, and sea ice: *Euphausia superba* and its changing environment, *BioScience*, 56(2).
- Nicol, S., J. Foster, and S. Kawaguchi (2012), The fishery for Antarctic krill - recent developments, *Fish and Fisheries*, 13(1), 30–40, doi:10.1111/j.1467-2979.2011.00406.x.
- Nicol, S., A. Worby, and R. Leaper (2008), Changes in the Antarctic sea ice ecosystem: Potential effects on krill and baleen whales, *Marine and Freshwater Research*, 59(5), 361, doi:10.1071/MF07161.
- Nicol, S., T. Pauly, N. L. Bindoff, S. Wright, D. Thiele, G. W. Hosie, P. G. Strutton, and E. Woehler (2000), Ocean circulation off East Antarctica affects ecosystem structure and sea-ice extent, *Nature*, 406(6795), 504–7, doi:10.1038/35020053.
- Nomura, D., H. Yoshikawa-Inoue, T. Toyota, and K. Shirasawa (2010), Effects of snow, snowmelting and refreezing processes on air – sea-ice CO₂ flux, *Journal of Glaciology*, 56(196), 262–270.
- Ohshima, K. I., Y. Fukamachi, G. D. Williams, S. Nihashi, F. Roquet, Y. Kitade, T. Tamura, D. Hirano, L. Herraiz-Borreguero, I. Field, M. Hindell, S. Aoki, and M. Wakatsuchi (2013), Antarctic Bottom Water production by intense sea-ice formation in the Cape Darnley polynya, *Nature Geoscience*, 6(3), 235–240, doi:10.1038/ngeo1738.
- Parkinson, C. L. (2002), Trends in the length of the Southern Ocean sea-ice season, 1979-99, *Annals of Glaciology*, 34(1), 435–440, doi:10.3189/172756402781817482.
- Perovich, D. K. (2011), The changing Arctic sea ice cover, *Oceanography*, 24(3), 162–173.
- Perovich, D. K., J. A. Richter-Menge, K. F. Jones, and B. Light (2008), Sunlight, water, and ice: Extreme Arctic sea ice melt during the summer of 2007, *Geophysical Research Letters*, 35, 1–4.
- Petrich, C., and H. Eicken (2010), Growth, structure and properties of sea ice, in *Sea Ice*, edited by

- D. Thomas and G. Dieckmann, pp. 23–78, Wiley-Blackwell.
- Pezza, A. B., H. A. Rashid, and I. Simmonds (2012), Climate links and recent extremes in Antarctic sea ice, high-latitude cyclones, Southern Annular Mode and ENSO, *Climate Dynamics*, **38**, 57–73, doi:10.1007/s00382-011-1044-y.
- Pritchard, H. D., S. R. M. Ligtenberg, H. A. Fricker, D. G. Vaughan, M. R. van den Broeke, and L. Padman (2012), Antarctic ice-sheet loss driven by basal melting of ice shelves, *Nature*, **484**(7395), 502–5, doi:10.1038/nature10968.
- Raphael, M. N., W. Hobbs and I. Wainer (2011), The effect of Antarctic sea ice on the Southern Hemisphere atmosphere during the southern summer, *Climate Dynamics*, **36**(7-8), 1403-1417.
- Renwick, J. A. (2002), Southern Hemisphere Circulation and Relations with Sea Ice and Sea Surface Temperature, *Journal of Climate*, **15**(21), 3058–3068, doi:10.1175/1520-0442(2002)015<3058:SHCARW>2.0.CO;2.
- Rintoul S.R. (1998). On the origin and influence of Adélie Land Bottom Water. In: S. Jacobs and R. Weiss (Eds.), *Ocean, Ice and Atmosphere: Interactions at the Antarctic Continental Margin*. AGU Antarctic Research Series 75, pp. 151-171. American Geophysical Union, Washington D.C.
- Rintoul, S. R., C. Hughes, and D. Olbers (2001), The Antarctic Circumpolar Current System, in *Ocean Circulation and Climate*, vol. 77, In: G. Siedler and J. A. Church (Eds.), pp. 271–302.
- Rintoul, S. R., S. Sokolov and R. A. Massom (2008), Rapid development and persistence of a massive Antarctic sea ice tongue, *Journal of Geophysical Research*, **113**, 1–16, doi:10.1029/2007JC004541.
- Rothrock, D. B. Percival, and M. Wensnahan (2008), The decline in Arctic sea-ice thickness: Separating the spatial, annual, and interannual variability in a quarter century of submarine data, *Journal of Geophysical Research*, **113**, 1–9.
- Russell, J. L., K. W. Dixon, A. Gnanadesikan, R. J. Stouffer, and J. R. Toggweiler (2006), The Southern Hemisphere westerlies in a warming world: Propping open the door to the deep ocean, *Journal of Climate*, **19**, 6382–6390.
- Rysgaard, S., R. N. Glud, M. K. Sejr, J. Bendtsen, and P. B. Christensen (2007), Inorganic carbon transport during sea ice growth and decay: A carbon pump in polar seas, *Journal of Geophysical Research*, **112**(C3), C03016, doi:10.1029/2006JC003572.
- Screen, J. A. and I. Simmonds (2010), Increasing fall-winter energy loss from the Arctic Ocean and its role in Arctic temperature amplification, *Geophysical Research Letters*, **37**, 1–5, doi:10.1029/2010GL044136.
- Sen-Gupta, A., A. Santoso, A. S. Taschetto, C. C. Ummenhofer, J. Trevena, and M. H. England (2009), Projected changes to the Southern Hemisphere ocean and sea ice in the IPCC AR4 climate models, *Journal of Climate*, **22**, 3047–3078, doi:10.1175/2008JCLI2827.1.
- Shadwick, E. H., S. R. Rintoul, B. Tilbrook, G. D. Williams, N. Young, A. D. Fraser, H. Marchant, J. Smith, and T. Tamura (2013), Glacier tongue calving reduced dense water formation and enhanced carbon uptake, *Geophysical Research Letters*, **40**, 904–909, doi:10.1002/grl.50178.
- Simmonds, I. (2003), Modes of atmospheric variability over the Southern Ocean, *Journal of Geophysical Research*, **108**(C4), doi:10.1029/2000JC000542.
- Simmonds, I., and I. Rudeva (2012), The great Arctic cyclone of August 2012, *Geophysical Research Letters*, **39**(23), L23709, doi:10.1029/2012GL054259.
- Simpkins, G.R., Ciasto, L.M. & England, M.H., 2013. Observed variations in multidecadal Antarctic sea ice trends during 1979-2012. *Geophysical Research Letters*, **40**(14), pp.3643–3648, doi:10.1002/grl.50715.
- Smetacek, V., and S. Nicol (2005), Polar ocean ecosystems in a changing world, *Nature*, **437**(7057),

362–8, doi:10.1038/nature04161.

- Smith, W. O., and J. C. Comiso (2008), Influence of sea ice on primary production in the Southern Ocean: A satellite perspective, *Journal of Geophysical Research*, 113(C5), C05S93, doi:10.1029/2007JC004251.
- Smith, W. O., and D. M. Nelson (1985), Phytoplankton bloom produced by a receding ice edge in the Ross Sea: Spatial coherence with the density field, *Science*, 227, 163–166.
- Stammerjohn, S., R. Massom, D. Rind, and D. Martinson (2012), Regions of rapid sea ice change: An inter-hemispheric seasonal comparison, *Geophysical Research Letters*, 39(6), 1–8, doi:10.1029/2012GL050874.
- Stammerjohn, S. E., D. G. Martinson, R. C. Smith, X. Yuan, and D. Rind (2008), Trends in Antarctic annual sea ice retreat and advance and their relation to El Niño – Southern Oscillation and Southern Annular Mode variability, *Journal of Geophysical Research*, 113, 1–20, doi:10.1029/2007JC004269.
- Stefels, J., M. Steinke, S. Turner, G. Malin, and S. Belviso (2007), Environmental constraints on the production and removal of the climatically active gas dimethylsulphide (DMS) and implications for ecosystem modelling, *Biogeochemistry*, 83(1-3), 245–275, doi:10.1007/s10533-007-9091-5.
- Stroeve, J., M. M. Holland, W. Meier, T. Scambos, and M. Serreze (2007), Arctic sea ice decline: Faster than forecast, *Geophysical Research Letters*, 34, 1–5, doi:10.1029/2007GL029703.
- Sturm, M., and R. A. Massom (2010), Snow and sea ice, in *Sea Ice*, edited by D. Thomas and G. Dieckmann, pp. 153–204, Wiley-Blackwell.
- Tamura, T., G. D. Williams, A. D. Fraser, and K. I. Ohshima. 2012. Potential regime shift in decreased sea ice production after the Mertz Glacier calving. *Nature Communications*, 3:826. doi:10.1038/ncomms1820.
- Thomas, D., and G. Dieckmann (Eds.) (2010), *Sea Ice*, 2nd ed., Blackwell Publishing Ltd.
- Thompson, D. W. J., S. Solomon, P. J. Kushner, M. H. England, K. M. Grise, and D. J. Karoly (2011), Signatures of the Antarctic ozone hole in Southern Hemisphere surface climate change, *Nature Geoscience*, 4(11), 741–749, doi:10.1038/ngeo1296.
- Toggweiler, J. R., and B. Samuels (1995), Effect of sea ice on the salinity of Antarctic bottom waters, *Journal of Physical Oceanography*, 25(9), 1980–1997.
- Trevena, A. J., and G. B. Jones (2006), Dimethylsulphide and dimethylsulphonioacetate in Antarctic sea ice and their release during sea ice melting, *Marine Chemistry*, 98(2-4), 210–222, doi:10.1016/j.marchem.2005.09.005.
- Trivelpiece, W. Z., J. T. Hinke, A. K. Miller, C. S. Reiss, S. G. Trivelpiece, and G. M. Watters (2011), Variability in krill biomass links harvesting and climate warming to penguin population changes in Antarctica, *Proceedings of the National Academy of Sciences of the United States of America*, 108(18), 7625–8, doi:10.1073/pnas.1016560108.
- Turner, J., and G. J. Marshall (2011), *Climate Change in the Polar Regions*, Cambridge University Press, Cambridge UK.
- Turner, J., and J. Overland (2009), Contrasting climate change in the two polar regions, *Polar Research*, 28(2), 146–164, doi:10.1111/j.1751-8369.2009.00128.x.
- Turner, J., T. J. Bracegirdle, T. Phillips, G. J. Marshall, and J. S. Hosking (2013), An initial assessment of Antarctic sea ice extent in the CMIP5 models, *Journal of Climate*, 26(5), 1473–1484, doi:10.1175/JCLI-D-12-00068.1.
- Turner, J., J. C. Comiso, G. J. Marshall, T. A. Lachlan-Cope, T. Bracegirdle, T. Maksym, M. P. Meredith, Z. Wang, and A. Orr (2009), Non-annular atmospheric circulation change induced by stratospheric ozone depletion and its role in the recent increase of Antarctic sea ice extent,

- Geophysical Research Letters*, 36, 1–5, doi:10.1029/2009GL037524.
- Vancoppenolle, M., K. M. Meiners, C. Michel, L. Bopp, F. Brabant, G. Carnat, B. Delille, D. Lannuzel, G. Madec, S. Moreau, J.-L. Tison and P. van der Merwe (2013) Role of sea ice in global biogeochemical cycles: emerging views and challenges. *Quaternary Science Reviews*, pp.1–24.
- van der Merwe, P., D. Lannuzel, A. R. Bowie, and K. M. Meiners (2011), High temporal resolution observations of spring fast ice melt and seawater iron enrichment in East Antarctica, *Journal of Geophysical Research*, 116(G3), G03017, doi:10.1029/2010JG001628.
- Williams, G. D. T. Maksym, C. Kunz, P. Kimball, H. Singh, J. Wilkinson, T. Lachlan-Cope, E. Trujillo, A. Steer, R. Massom, K. Meiners, P. Heil, J. Lieser, K. Leonard, and C. Murphy (2013), Beyond point measurements: Sea ice floes characterized in 3-D, *Eos, Transactions American Geophysical Union*, 94(7), 69–70, doi:10.1002/2013EO070002.
- Worby, A. P., R. A. Massom, I. Allison, V. I. Lytle, and P. Heil (1998), East Antarctic Sea Ice: A Review of Its Structure, Properties and Drift. In *Antarctic Sea Ice: Physical Processes, Interactions and Variability*, ed. O. Jeffries, 41–67. Antarctic . Washington, D.C.: AGU. doi:10.1029/AR074p0041.
- Worby, A. P., C. A. Geiger, M. J. Paget, M. L. van Woert, S. F. Ackley, and T. L. DeLiberty (2008), Thickness distribution of Antarctic sea ice, *Journal of Geophysical Research*, 113, 1–14.
- Wu, X., W. F. Budd, and T. H. Jacka (1999), Simulations of Southern Hemisphere warming and Antarctic sea-ice changes using global climate models, *Annals Of Glaciology*, 29(1), 61–65, doi:10.3189/172756499781821517.
- Yi, D., H. J. Zwally, and J. W. Robbins (2011), ICESat observations of seasonal and interannual variations of sea-ice freeboard and estimated thickness in the Weddell Sea, Antarctica (2003–2009), *Annals of Glaciology*, 52(57), 43–51.
- Yuan, X., and C. Li (2008), Climate modes in southern high latitudes and their impacts on Antarctic sea ice, *Journal of Geophysical Research*, 113(C6), C06S91, doi:10.1029/2006JC004067.
- Zhang, J. (2007), Increasing Antarctic sea ice under warming atmospheric and oceanic conditions, *Journal of Climate*, 20, 2515–2529, doi:10.1175/JCLI4136.1.
- Zwally, H. J., D. Yi, R. Kwok, and Y. Zhao (2008), ICESat measurements of sea ice freeboard and estimates of sea ice thickness in the Weddell Sea, *Journal of Geophysical Research*, 113, 1–17



ANTARCTIC CLIMATE
& ECOSYSTEMS CRC